Literature Review: Offline Reinforcement Learning
Version 2: Parts A and B

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Part A: Research Papers

Papers 1–8 carry over from Version 1; Papers 9–18 are new in Version 2. Every work is described with the same four themes, of which each paper names its three most central: distribution shift, the conservatism–improvement trade-off, uncertainty quantification and calibration, and theory vs. practice.

Paper 1: Conservative Q-Learning for Offline Reinforcement Learning (CQL)

CQL trains the critic to be pessimistic about actions the dataset does not contain, so the policy is less drawn to accidentally inflated scores.

1. **What is the citation?** Kumar, A., Zhou, A., Tucker, G., & Levine, S. (2020). Conservative Q-learning for offline reinforcement learning. *Advances in Neural Information Processing Systems, 33*, 1179–1191. Version reviewed: arXiv:2006.04779v3. `https://doi.org/10.48550/arXiv.2006.04779`

2. **What is the research area?** The area is machine learning, specifically offline reinforcement learning (RL): learning a policy from a fixed log of past decisions with no further trial and error. CQL belongs to the "pessimistic critic" family, which lowers the scores of actions outside the data instead of restricting the policy. This distinguishes it from policy-constraint methods (BEAR, BRAC) and ensemble/uncertainty methods.

3. **What is the research question?** Can a critic trained to be systematically pessimistic about unseen actions produce a policy whose predicted value is a guaranteed floor on its true performance? Does that beat methods that keep the policy near the recorded behavior?

4. **What is the significance of the question?** Offline RL is most valuable where live experimentation is expensive or dangerous (robotics, medicine, recommendation) but large logs already exist. Standard algorithms often fail there because the critic's guesses about unseen actions are never corrected and the policy is drawn to the inflated ones. The authors describe earlier fixes (modeling the behavior policy, uncertainty penalties) as limited. A cheap, principled fix could let RL reuse logged data the way supervised learning reuses datasets.

5. **What are the contributions reported in this paper?** The novel idea is a simple addition to the critic's training objective. It pushes down the scores of actions the critic likes too much

and props up those of actions in the data, so the net penalty lands mainly on unseen actions. The paper provides two ready-to-use variants, CQL($\mathcal{H}$) and CQL(ρ), and a roughly twenty-line modification of two standard algorithms with an automatic scheme for setting the caution strength α. It also proves, under stated assumptions and for large enough α, that the learned value undercuts the true value and that, with high probability, the policy is at most a bounded amount worse than the one that collected the data ("safe improvement").

6. **What is the research method?** The paper combines algorithm design, mathematical analysis (mostly for simplified tabular cases, with extensions), and broad benchmarking against established methods on the D4RL suite (simulated locomotion, mazes, dexterous-hand and kitchen tasks) and offline Atari. Diagnostics check whether predicted values stay below true returns, and ablations isolate key design choices.

7. **What are the results?** On datasets recorded by a single behavior, CQL is close to the best prior method. On the mixed-source datasets the authors consider more realistic, it usually wins by large margins (e.g., 91.1–110.5 on the three random-expert locomotion tasks, where the best competitors score 53.0, 2.7, and 11.1). Only the CQL variants score above zero on the hardest mazes. As designed, its predicted values sit below true returns on the three datasets checked, while the baselines compared there overestimate.

8. **What are the impacts and limitations of the contributions?** CQL is a comparison baseline in many later papers reviewed here (e.g., TD3+BC, IQL, EDAC, ReBRAC) and showed pessimism can be added with a small code change, without modeling the behavior that produced the data. Its guarantees, however, cover simplified settings rather than the deep networks actually used. The caution knob α (or its auto-tuning threshold) was set differently across domains, and there is no established way to decide when to stop training without trying the policy live. One global α also applies the same strength to well-covered and poorly covered situations.

9. **How were the results evaluated or validated?** The paper relies on mathematical proofs in appendices, D4RL scores averaged over four runs, direct comparisons of predicted values with actual returns, and ablations. The evidence is broad but statistically thin: tables of means without spreads, some baselines copied rather than rerun, and a few inconsistent tables.

10. **What are the ethical implications? How are they addressed?** The reviewed version contains no broader-impact discussion. The "safe improvement" guarantee is only relative to the logging policy, probabilistic, and assumption-bound, so the word *safe* could invite misplaced trust. A policy kept near the data also keeps the data's biases. All validation is simulated, and compute cost is unreported. Conservatism is an indirect step toward safer learning, but bias, deployment safeguards, and societal consequences go undiscussed.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: scores for unseen actions cannot be corrected offline. *The conservatism–improvement trade-off*: pessimism must be dosed, here via α. *Theory vs. practice*: the guarantees are not proved for the deep networks actually used.

Connection to the offline RL problem space and BCA: CQL exemplifies the "one global pessimism knob" paradigm of the problem space. Applied to CQL, BCA (Bayesian Conformal Aggregation) keeps the critic-side idea but is designed to scale the conservative gap per sample by a calibrated width, on top of the single α. CQL's appendix sketches a loosely related count-dependent penalty for the tabular case.

Paper 2: A Minimalist Approach to Offline Reinforcement Learning (TD3+BC)

TD3+BC shows that a standard online algorithm with one added "stay near the recorded actions" penalty matches more complicated offline methods on the D4RL locomotion tasks at a fraction of the compute.

1. **What is the citation?** Fujimoto, S., & Gu, S. S. (2021). A minimalist approach to offline reinforcement learning. *Advances in Neural Information Processing Systems, 34*, 20132–20145. Version reviewed: arXiv:2106.06860v2. `https://doi.org/10.48550/arXiv.2106.06860`

2. **What is the research area?** The area is offline RL for continuous control (robot-like tasks with smooth action spaces). TD3+BC belongs to the behavior-regularization family, which fights inflated scores for unseen actions by keeping the policy close to the recorded behavior.

3. **What is the research question?** Can a standard online algorithm, changed as little as possible, reach state-of-the-art offline performance, and how much of the extra machinery in recent offline methods is necessary?

4. **What is the significance of the question?** Every extra component (a learned model of the recorded behavior, pre-training phases, architecture changes) is a choice that normally must be validated by trying policies, which requires the interaction offline RL is supposed to avoid. A minimal method helps isolate what works, raises the bar elaborate methods must clear, and stays cheap as datasets grow.

5. **What are the contributions reported in this paper?** The paper makes three contributions. It audits two leading methods (CQL and Fisher-BRC), showing they owed much of their scores to poorly justified engineering tweaks, not just their stated ideas. It proposes TD3+BC: one penalty term in the actor's objective ("score high, but stay close to the recorded action") plus standardized inputs, with one balance setting reused across all tasks. It also gives an example of careful practice: unified reruns of the two strongest competitors, wall-clock timings, a demonstration that the offline policies tested are unstable across episodes, and public code.

6. **What is the research method?** The study combines critical analysis with experimentation rather than new mathematics. The actor maximizes $\lambda Q(s, \pi(s)) - (\pi(s) - a)^2$ averaged over recorded state-action pairs. The balance weight λ is automatically rescaled by the typical size of the scores, so one value ($\alpha = 2.5$) is used for every dataset. Experiments cover fifteen standard locomotion datasets with five random seeds, plus ablations removing each ingredient.

7. **What are the results?** Summed over the fifteen tasks, TD3+BC essentially ties the best competitor (979.3 vs. 974.6) and beats its rerun of CQL by a wide margin (764.3). It trains in 39 minutes versus over four hours for CQL. Ablations confirm both the RL term and the penalty are needed on most datasets; normalization gives a smaller boost. TD3+BC does poorly on some datasets heavy in poor behavior (28.8 vs. CQL's 100.3 on a half-random, half-expert Walker2d mix) and on the hardest mazes.

8. **What are the impacts and limitations of the contributions?** TD3+BC became what ReBRAC (Paper 4) calls "a de-facto standard baseline to be compared against". It also urged the community to report stability, not just averages. The paper offers no theory. A uniform pull toward recorded actions cannot tell good recordings from bad ones and can average over different behaviors. The diagnosed instability is left unsolved, and every sample receives the same constraint strength regardless of how well the data cover it.

9. **How were the results evaluated or validated?** The paper uses five seeds and ten test episodes per checkpoint, and averages scores over the final checkpoints rather than the luckiest one. The two strongest competitors (CQL and Fisher-BRC) were rerun from their authors' code under one protocol. There are no statistical tests or confidence intervals and only five seeds despite high documented variance. The tie with Fisher-BRC supports parity rather than superiority.

10. **What are the ethical implications? How are they addressed?** A short statement notes that learning from logged data inherits its biases, warns of misuse (e.g., more habit-forming recommendation), and argues the main effect is wider access. Arguably the paper's most useful safety contribution is documenting that offline policies can fail badly on individual episodes. No mitigations are proposed, and all evidence is simulated.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: inflated scores for unseen actions are fought by keeping the policy close to the recorded behavior. *The conservatism–improvement trade-off*: TD3+BC uses one uniform pull toward the data, with a single balance setting, and does poorly on some low-quality datasets and the hardest mazes. *Theory vs. practice*: a few changed lines matched prior methods on locomotion tasks, and engineering, not just ideas, drove part of two prior methods' scores.

Connection to the offline RL problem space and BCA: TD3+BC is the minimalist pole of the problem space, with one uniform constraint for every sample. Its successor (ReBRAC, Paper 4) found in its ablations that removing this constraint caused the largest overall drop. BCA is designed to strengthen this term per sample by a bounded, calibrated factor while keeping the global weight.

Paper 3: Offline Reinforcement Learning with Implicit Q-Learning (IQL)

IQL learns values using only the actions in the dataset, never asking "what if" about unseen actions, yet can still stitch together good behavior from suboptimal pieces.

1. **What is the citation?** Kostrikov, I., Nair, A., & Levine, S. (2022). Offline reinforcement learning with implicit Q-learning. In *International Conference on Learning Representations (ICLR 2022)*. Version reviewed: arXiv:2110.06169v1. `https://doi.org/10.48550/arXiv.2110.06169`

2. **What is the research area?** The area is offline RL. IQL takes a third path beside constraining the policy (TD3+BC) and penalizing the critic (CQL): *in-sample* learning, in which value networks are evaluated only on recorded actions. The policy is extracted afterward by imitation weighted toward better-than-expected actions (AWR).

3. **What is the research question?** Can an offline algorithm still plan ahead by chaining many steps of value backups toward optimal behavior while its value functions are trained only on logged actions, so that unseen actions are never scored?

4. **What is the significance of the question?** Asking the critic about unseen actions is a main source of offline error, and the prior fixes the paper discusses compromise between safety and improvement. One-step in-sample alternatives avoid the problem but do poorly when fragments of good behavior must be stitched together, as when a maze solution must be assembled from partial trajectories. Removing the problem while keeping multi-step planning would ease the field's central tension, cheaply.

5. **What are the contributions reported in this paper?** The novel core is "implicit" value learning. A state-value network trained with an asymmetric loss (an expectile loss, whose level acts as an optimism dial) estimates the best value achievable within the data. A second network then learns action values one step further, so the pair plans ahead without scoring an unseen action. The paper also proves, for idealized exact solutions, that the fully optimistic limit of the dial recovers the best in-data value. Its other contributions are policy extraction by weighted imitation, a correction of earlier unfair benchmark comparisons, an offline-to-online fine-tuning study, and fast public code.

6. **What is the research method?** The paper combines algorithm design and a short limiting argument with benchmarks: a toy maze built to require stitching, and standard locomotion, maze, hand-manipulation, and kitchen suites against up to seven baselines. It also ablates the optimism dial and runs a fine-tuning study in which offline training is followed by live interaction.

7. **What are the results?** On locomotion, IQL roughly ties the best prior method (692.4 vs. 698.5). On the maze tasks that demand stitching it leads on five of six (378.0 vs. 303.6 total), scoring 39.6 and 47.5 on the largest mazes, where the strongest competitor manages about 15. Training takes about 20 minutes. After live fine-tuning IQL improves further (370 to 474 on mazes); CQL also improves in total (152 to 231) but falls to zero on one medium maze. IQL still loses individual tasks, for example to filtered imitation on two medium-expert locomotion datasets.

8. **What are the impacts and limitations of the contributions?** IQL showed that in-sample learning need not give up long-horizon planning. Because its value learning never depends on the policy, the extraction step could in principle be swapped, and can be continued online. However, its two knobs are set per domain, and the tables report means without spreads. Continuous states usually carry only one recorded action, so the "best in-data value" relies on the network generalizing across similar states. Its fixed weighting also amplifies trustworthy and noisy advantages identically.

9. **How were the results evaluated or validated?** The paper reports standard benchmark averages (ten seeds on locomotion, 100 episodes on mazes), a corrected all-one-version comparison, and one ablation. Baseline numbers come from mixed sources (one competitor's maze scores could not be reproduced and were copied). The result tables have no spreads, no procedure is stated for choosing per-domain settings, and only the optimism dial is ablated.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics discussion. Because the policy is built by reweighting recorded behavior, biases in the log are inherited and can be amplified whenever they scored well. The deliberate optimism of the upper expectile may be risky in safety-critical use, and the fine-tuning recipe involves live exploration that would carry physical risk on hardware. Low compute cost is a modest positive.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: IQL sidesteps it during value learning, since training never queries unseen actions. *The conservatism–improvement trade-off*: the expectile level sets how optimistic the values are, and the temperature of the imitation weights sets how closely the policy follows the data. *Theory vs. practice*: the proof concerns the fully optimistic limit of the dial, while in practice its two knobs are set per domain and the "best in-data value" relies on the network generalizing across similar states.

Connection to the offline RL problem space and BCA: IQL is one of BCA's four host algorithms. Its imitation weight is the term BCA is designed to temper per sample with a calibrated width,

while keeping IQL's global temperature.

Paper 4: Revisiting the Minimalist Approach to Offline Reinforcement Learning (ReBRAC)

ReBRAC shows that giving the simple TD3+BC baseline the quiet engineering upgrades of newer methods, including deeper networks, normalization, and separate penalties, is enough to match or beat more complex ensemble-free algorithms on average.

1. **What is the citation?** Tarasov, D., Kurenkov, V., Nikulin, A., & Kolesnikov, S. (2023). Revisiting the minimalist approach to offline reinforcement learning. *Advances in Neural Information Processing Systems, 36*, 11592–11620. Version reviewed: arXiv:2305.09836v2. `https://doi.org/10.48550/arXiv.2305.09836`

2. **What is the research area?** The area is offline RL, in the behavior-regularization lineage (BRAC, TD3+BC). The paper deliberately proposes no new principle and studies systematically which "incidental" design choices bundled with new algorithms improve an established baseline.

3. **What is the research question?** How far can a minimalist baseline be improved purely by adopting the small design choices that recent methods shipped quietly alongside their headline ideas, and which of them account for the gain? The choices are deeper networks, layer normalization, bigger batches, separate penalty weights for actor and critic, and a longer planning horizon.

4. **What is the significance of the question?** If gains credited to new algorithms actually come from quiet engineering, the field mis-attributes progress and practitioners cannot tell what is worth adopting. Establishing how strong a cheap baseline can become raises the bar new methods should clear and gives practitioners a fast default. Some elaborate alternatives were expensive (one needed as many as 500 critic networks).

5. **What are the contributions reported in this paper?** The paper audits which of five design choices seven recent algorithms adopt and introduces ReBRAC: TD3+BC plus five upgrades, including separate "stay near the data" penalties on the actor and on the critic. It evaluates ReBRAC on 51 datasets spanning state-based and image-based benchmarks plus an offline-to-online study. Large-scale ablations identify what carries the gains, and a full reproducibility package supplies code, per-dataset settings, and compute tables.

6. **What is the research method?** The method is design by composition, then large-scale benchmarking and ablation, with no new theory. For the state-based tasks, settings are tuned on four random seeds, and the chosen configuration is re-evaluated on ten fresh seeds using final-checkpoint reporting, a fairer protocol than reporting lucky peaks. Diagnostic runs remove or add one upgrade at a time.

7. **What are the results?** On average, ReBRAC is the best of the compared ensemble-free methods on mazes (76.8) and hand-manipulation tasks (58.6) and a close second on locomotion (81.2 vs. SAC-RND's 82.6). It fine-tunes online from 55.0 to 91.1 with the lowest average regret of the methods tested. The ablations are arguably the more informative finding. Removing the actor's stay-near-the-data penalty makes the method collapse (72.2 to 11.4), removing critic layer normalization halves the score, and the critic-side penalty adds almost nothing on average. No single tested upgrade alone lifts plain TD3+BC to ReBRAC's level.

8. **What are the impacts and limitations of the contributions?** ReBRAC supplied a stronger cheap baseline (Papers 6 and 8 here compare against it) and practical guidance from its ablations: the actor penalty and critic normalization are essential, and the planning horizon matters when rewards are sparse. However, its two penalty weights are still re-tuned per dataset by trying policies live (best values span two orders of magnitude), and constraint strength is uniform across samples *within* a dataset. Most tasks recorded by humans remain unsolved, there is no theory, and everything is simulated.

9. **How were the results evaluated or validated?** The evaluation is fairly rigorous, with separate tuning and reporting seeds, final-checkpoint reporting, modern aggregate statistics, and a control showing the gains come from the design choices rather than implementation differences. Still, model selection uses real environment returns, which a true offline user lacks, and copied baselines come from other papers' protocols, which may be more lenient.

10. **What are the ethical implications? How are they addressed?** Ethics are not discussed beyond scientific scope. Implicitly, staying near logged behavior inherits its biases and unsafe habits, and tuning by live trial runs contradicts the offline premise in risky domains. Performance also dipped on one task during live fine-tuning, which outside simulation would mean worse behavior in operation. Compute (roughly 18,500 GPU-hours) is disclosed but not discussed.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: the safeguard against unsupported actions is the actor's stay-near-the-data penalty, and the method collapses without it (72.2 to 11.4). *The conservatism–improvement trade-off*: the penalty weights are still re-tuned per dataset by live evaluation, and constraint strength is uniform across samples within a dataset. *Theory vs. practice*: incidental choices may account for much reported "algorithmic" advance, and some baselines are re-tuned under the same protocol.

Connection to the offline RL problem space and BCA: ReBRAC's ablations identify the actor's stay-near-the-data term as the lever that matters most, but its strength remains a single scalar re-tuned per dataset by live evaluation. BCA's calibrated weighting targets the same lever, dosed per sample within a bounded range on top of the host's tuned scalar.

Paper 5: Importance Weighted Actor-Critic for Optimal Conservative Offline Reinforcement Learning (A-Crab)

A-Crab proves that a small change to the critic's error penalty, weighting each sample's error instead of squaring it, turns a slow-but-practical method into one that attains the optimal statistical rate under its assumptions.

1. **What is the citation?** Zhu, H., Rashidinejad, P., & Jiao, J. (2023). Importance weighted actor-critic for optimal conservative offline reinforcement learning. *Advances in Neural Information Processing Systems, 36*, 49579–49602. Version reviewed: arXiv:2301.12714v2. `https://doi.org/10.48550/arXiv.2301.12714`

2. **What is the research area?** The area is theoretical offline RL with general function approximation, meaning its guarantees are not tied to toy problems. The paper connects adversarial actor-critic methods (ATAC), where the critic plays devil's advocate against the policy, with importance weighting, which corrects for how rarely some behavior appears in the data.

3. **What is the research question?** Can one practical algorithm achieve the statistically optimal learning speed (error shrinking like $1/\sqrt{N}$ as the data grow) under a weak, averaged assumption about how well the data cover alternative behaviors? Prior practical methods had slower proven rates or needed stronger, worst-case coverage.

4. **What is the significance of the question?** A guarantee is only as useful as its assumptions. Real logs (medical records, robot traces) rarely cover every possible policy, and worst-case coverage assumptions collapse when a single rarely logged action makes a likelihood ratio enormous. A practical method with the optimal rate under weak, averaged coverage would narrow the gap between offline RL theory and practice.

5. **What are the contributions reported in this paper?** The key step replaces the squared Bellman-error penalty with the largest *importance-weighted average* error (a Bellman error is the critic's disagreement with its own one-step lookahead). The weights emphasize the samples that matter for the policies being compared. The paper adds an optimal-rate proof that drops the strong "completeness" closure assumption predecessors needed, and a weaker coverage measure. It guarantees a policy no worse than the logging policy, up to a vanishing error, across a wide range of penalty weights. It also proves the old squared penalty fails to reach the optimal rate on some problems and provides a practical variant with public code.

6. **What is the research method?** Mathematics comes first: the paper analyzes a game in which the actor tries to beat the logged behavior while the critic minimizes that improvement plus the weighted error penalty, using finite-sample concentration tools. A tractable approximation then restricts the weights to a boxed range, yielding a closed-form penalty, and is benchmarked against ATAC on twelve simulated locomotion datasets.

7. **What are the results?** In theory, under its assumptions, A-Crab attains the optimal $1/\sqrt{N}$ rate against any comparator policy the data cover, improving on the slower rates of two tractable predecessors ($1/N^{1/3}$ and $1/N^{1/6}$). Under the same assumptions, the learned policy is guaranteed to do no worse than the logging policy, up to a vanishing error, across a wide range of penalty settings. In practice it usually trains more stably than ATAC and scores better on several datasets (e.g., about 110 vs. a noisy 90–100 on one locomotion task), though results appear only as training curves.

8. **What are the impacts and limitations of the contributions?** The main impact is conceptual: in the paper's analysis, the *form* of the error penalty (linear and sample-weighted rather than squared) is what separates a tractable-but-slow method from a rate-optimal one. However, the assumptions (finite action sets, idealized actor updates, well-behaved weights) are far from deep-learning practice, and the benchmarked algorithm departs from the analyzed one. The per-sample weighting also reduces in practice to a closed-form penalty with a single grid-searched constant per dataset.

9. **How were the results evaluated or validated?** The theorems have full appendix proofs built from standard statistical tools. The empirical side is thin: it shows only training curves against a single baseline, with no final-score table or statistical tests. No results are shown for the new penalty without the squared one, and the theoretical rate improvement is not checked numerically.

10. **What are the ethical implications? How are they addressed?** There is no ethics discussion. Optimal-rate language could create false confidence for clinical or robotic use. The guarantees rest on assumptions that are hard to verify from logged data, concern average return

rather than per-state safety, and do not cover the benchmarked version. Because the method competes only with behaviors the data cover, logging bias can carry over into the learned policy. Public code is the one clear positive.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: real logs rarely cover every policy, so guarantees are sought under partial coverage, with optimal rates against covered behaviors only. *The conservatism–improvement trade-off*: weighting each sample's error instead of squaring it gives the optimal rate in the paper's analysis. The analysis also guarantees a policy no worse than the logging policy (up to a vanishing error) across a wide range of penalty weights. *Theory vs. practice*: the tested algorithm is not the analyzed one, and the per-sample weighting reduces in practice to a closed-form penalty with one grid-searched constant per dataset.

Connection to the offline RL problem space and BCA: A-Crab's per-sample reweighting of prediction errors is a loose theoretical cousin of BCA's calibrated widths. The paper proves its importance weighting attains the optimal statistical rate under its assumptions, a result that does not carry over to BCA. Its practical reduction to one searched constant per dataset is the kind of gap BCA is designed to target.

Paper 6: A Clean Slate for Offline RL (Unifloral)

This paper rebuilds twelve existing algorithms in one fast codebase and grades ten of them by how well each scores for a given amount of live testing spent on tuning, a cost earlier papers rarely stated.

1. **What is the citation?** Jackson, M. T., Berdica, U., Liesen, J., Whiteson, S., & Foerster, J. N. (2025). A clean slate for offline reinforcement learning. *Advances in Neural Information Processing Systems, 38*. Version reviewed: arXiv:2504.11453v1. `https://doi.org/10.48550/arXiv.2504.11453`

2. **What is the research area?** The area is offline RL methodology, evaluation, and software rather than a new learning rule. Its problem family is hyperparameter and policy selection under limited online access, together with fair benchmarking. This places it beside earlier unification and reimplementation efforts (Rainbow, CORL).

3. **What is the research question?** How should offline RL methods be defined, implemented, and compared so that reported scores honestly reflect the live interaction spent tuning them? Can one cleanly implemented design space, containing prior algorithms as settings, be used to discover better ones?

4. **What is the significance of the question?** Offline RL's selling point, learning without touching the real world, is undermined if published scores quietly depend on many live test episodes used to pick settings. Numbers from different papers then mix algorithmic quality with tuning effort, and the two are hard to tell apart. By 2025 the field had many competing methods, copied baselines, and inconsistent codebases.

5. **What are the contributions reported in this paper?** The paper makes four contributions. It distinguishes four offline RL settings by when live interaction is allowed, and defines "a method" as an algorithm plus one fixed range of settings used on every dataset. Its evaluation protocol simulates tuning with a bandit (a strategy for choosing which trained policy to test next), so performance is plotted against the live-testing budget. It draws a family tree of prior methods,

reimplements twelve of them, and builds Unifloral, one configurable implementation covering most of the tree (it omits CQL). It creates two new algorithms (TD3-AWR and MoBRAC) by mixing settings, with no new code. All code is public.

6. **What is the research method?** The method is conceptual analysis, protocol and framework construction, then benchmarking, with no new theory. For each algorithm–dataset pair, 20 policies are trained with settings sampled from the original papers' ranges and their episode returns stored. Then 500 simulated tuning sessions select among them under a stated budget. The study uses nine datasets, with timing runs on one GPU.

7. **What are the results?** None of the ten prior methods tested dominates. ReBRAC leads on five of nine datasets at some budget, IQL on four, and model-based methods never rise above sixth outside locomotion. The reimplementations are compact (59 changed lines turn TD3+BC into ReBRAC) and faster, in some cases dramatically (CQL trains in about 10 minutes vs. over three hours in a popular library). TD3-AWR beats each of its parent methods on most datasets (ReBRAC on six of nine, IQL on seven). With only a few noisy test episodes, selection can be fooled by erratic "distractor" policies (shown for ReBRAC on one dataset).

8. **What are the impacts and limitations of the contributions?** The paper offers the field a common yardstick and toolkit: hidden tuning becomes an explicit, plotted cost, and component-level ideas can be tested by changing settings rather than writing new code. However, only the pre-deployment setting is implemented, rankings depend on the original papers' tuning ranges, and everything uses one benchmark suite. Some fixed configuration choices also appear to degrade certain algorithms relative to their originals, and the two headline average speedups are not recoverable from the reported data.

9. **How were the results evaluated or validated?** The evaluation uses stored-episode tuning simulations with confidence intervals, a locomotion reproduction table said to match an earlier library's scores (reference numbers not shown), timing runs, and a distractor-policy analysis over 100,000 orderings. There are no formal statistical tests, and the intervals reflect selection randomness over one pool of trained policies rather than retraining. Some protocol details go unreported, and no reported check confirms the unified implementation matches each standalone file.

10. **What are the ethical implications? How are they addressed?** There is no dedicated ethics section, but the paper can be read as a research-integrity intervention: undisclosed live tuning can inflate deployability claims, and pricing that budget makes the inflation visible. It leaves unaddressed that selecting policies by average return ignores worst-case outcomes in the safety-critical domains it cites, and that distractor policies are the kind a careless real selection process could deploy. Every benchmark is also simulated. The efficiency gains could widen access for smaller labs.

11. **What themes are suggested by this research? (minimum 3)** *The conservatism–improvement trade-off*: many published methods can be written as settings of one implementation, and the strongest methods in its tests still tie the policy to the data. *Uncertainty quantification and calibration*: policy selection is itself uncertain. With only a few noisy test episodes it can be fooled by "distractor" policies (shown for one algorithm on one dataset), and the main budget curves have 95% confidence intervals over simulated tuning sessions. *Theory vs. practice*: scores that depend on live tuning undermine offline RL's promise of learning without live interaction. Here tuning is priced in episodes, in one codebase, with budget curves instead of headline numbers.

Connection to the offline RL problem space and BCA: Unifloral is the baseline reference bundled with BCA's code, and its budget-aware protocol is a suitable measuring stick for BCA's aim. If calibrated per-sample caution reduced the search over global knobs, the payoff would appear as better scores at small live-testing budgets. That remains untested.

Paper 7: Quantile Q-Learning: Revisiting Offline Extreme Q-Learning with Quantile Regression (QQL)

QQL replaces a sensitive hand-tuned knob in a recent offline algorithm, a single "optimism temperature" per dataset, with a per-state scale learned from the data.

1. **What is the citation?** Gao, X., Li, S., Cai, Y., & Yu, W. (2026). Quantile Q-learning: Revisiting offline extreme Q-learning with quantile regression. *Transactions on Machine Learning Research*. Version reviewed: arXiv:2511.11973v2. `https://doi.org/10.48550/arXiv.2511.11973`

2. **What is the research area?** The paper is in offline RL, downstream of the in-sample family (IQL, XQL), which learns values only on recorded actions and extracts the policy by weighted imitation. It addresses temperature selection, or how optimistic the value estimates should be, by borrowing quantile regression (learning specific percentiles rather than averages) from distributional RL.

3. **What is the research question?** Can an XQL-style algorithm stay stable and competitive across many domains with one fixed configuration if its temperature is estimated from the data per state instead of tuned per dataset?

4. **What is the significance of the question?** Per-dataset tuning normally means trying candidate policies live, the interaction offline RL is meant to avoid. XQL's temperature is a clear case, with reproduced values spanning 0.5 to 10 across datasets. It feeds both value learning and the policy's imitation weights, so a bad choice can destabilize both. A learned per-state scale would remove that knob.

5. **What are the contributions reported in this paper?** The paper first shows that, under an assumed noise model, the temperature equals a constant times the gap between two specific percentiles of the action-value distribution, so ordinary quantile regression can learn it without hand-tuning. It then builds QQL from two percentile value networks, a bias-corrected value target, and a policy update whose optimism automatically varies by state. It adds a cautious generalization term for unseen actions and a broad study (19 datasets from D4RL and NeoRL2, ablations, and checks of the noise model).

6. **What is the research method?** The paper derives its result under stated assumptions, then designs and benchmarks an algorithm. All settings are fixed across datasets, while baselines are allowed their tuned values. Further studies sweep the two main remaining knobs, test the noise model on synthetic data, and fit distributions to real prediction errors.

7. **What are the results?** With one fixed configuration, QQL posts the best average on 10 of 17 standard datasets. When XQL must also share one temperature, QQL wins clearly (locomotion 85.6 vs. 73.6; mazes 84.9 vs. 67.3). Its training curves are also smoother than those of a stabilized XQL variant on the four datasets plotted. It is weak on both Hammer manipulation datasets and on the one safety-constrained task, where it scores below the dataset's own average. Compute is modest (about four hours per run, 1.6× its parent).

8. **What are the impacts and limitations of the contributions?** QQL is evidence that XQL's temperature can be estimated per state from data, and its percentile-gap scale is a reusable idea. The assumed noise model appears to fit the reported residuals only loosely; they look heavier-tailed on the four datasets analyzed. The two learned percentiles can cross, which is patched by hand, and the learned scale carries no coverage guarantee. Some ablation claims overstate the figures, and no code link is given.

9. **How were the results evaluated or validated?** The evaluation uses five-run averages (three-run on NeoRL2), ablations of both new components, knob sweeps, a synthetic test, and an error-distribution analysis. The benchmark scores have no confidence intervals or significance tests, and baselines come from mixed sources. The choice of shared temperature for the head-to-head comparison is unexplained. The synthetic test injects exactly the noise it then detects, and a few numbers conflict internally.

10. **What are the ethical implications? How are they addressed?** The paper discusses none. Its claim that fixed settings make the method practical for consequential use rests entirely on simulation. The only safety-focused task is where it scores below the data, and constraint violations are not reported. A misestimated scale in sparsely covered states can mean *less* caution where mistakes may cost most, and logging bias is inherited. Fixed settings could reduce risky live trial runs, which is the paper's implicit safety benefit.

11. **What themes are suggested by this research? (minimum 3)** *The conservatism–improvement trade-off*: optimism is set per state instead of by one temperature, and wider value spread across recorded actions triggers stronger caution. *Uncertainty quantification and calibration*: the learned scale is a modeled quantity with no coverage guarantee. *Theory vs. practice*: hyperparameter sensitivity is arguably relocated rather than removed, since tuning moves from a global knob into a quantity learned under a noise model that appears to fit the reported residuals only loosely. That quantity's two controls look insensitive on the three datasets swept.

Connection to the offline RL problem space and BCA: QQL is a close non-conformal precedent for BCA: its per-state scale acts on the same imitation weight that BCA modulates in its IQL host. QQL's scale is learned under a parametric noise model with no guarantee. BCA's conformally calibrated width is designed as an alternative, motivated in part by the heavy-tailed residuals QQL reports, though its coverage concerns held-out residuals, not the policy.

Paper 8: UNIQ: Conformal Calibration for Adaptive Conservatism in Offline Reinforcement Learning (UNIQ)

UNIQ gives a standard offline learner (IQL) a low-memory, per-state caution dial driven by conformally rescaled ensemble disagreement, meant to give more caution where data are sparse and less where they are dense.

1. **What is the citation?** Upadhyay, A. (2026, July 11). *UNIQ: Conformal calibration for adaptive conservatism in offline reinforcement learning* [Poster presentation]. ICML 2026 Workshop on Decision-Making from Offline Datasets to Online Adaptation: Black-Box Optimization to Reinforcement Learning, Seoul, South Korea. Version reviewed: arXiv:2606.07592v1. `https://doi.org/10.48550/arXiv.2606.07592`

2. **What is the research area?** The paper is in offline RL and builds on IQL. The problem family is pessimism under distribution shift: how much to distrust value estimates in poorly covered situations. Unlike global-knob methods (CQL, IQL, TD3+BC), large-ensemble methods (SAC-N, EDAC; Paper 17), or learned-model penalties (MOPO), UNIQ pairs a small ensemble of value heads with conformal calibration. In UNIQ, this statistics recipe turns held-out prediction errors into a scale factor that makes the uncertainty signal unit-free.

3. **What is the research question?** Can IQL's global optimism setting be replaced by a per-state setting driven by conformally normalized ensemble disagreement, with tighter caution where data are sparse and looser where they are dense, without the memory cost of large ensembles?

4. **What is the significance of the question?** The best setting of most mainstream caution knobs changes per dataset and is normally found by trying policies live, an option unavailable offline. Raw ensemble disagreement, the usual uncertainty signal, has no common scale across tasks, so fixed penalties transfer poorly. A calibrated, lightweight, per-state rule would cut both live tuning and hardware demands.

5. **What are the contributions reported in this paper?** The main contribution is a three-part mechanism. A small ensemble of value heads, trained at three optimism levels, estimates uncertainty from its spread. A conformal step measures held-out prediction errors and converts raw spread into a unit-free ratio. A controller maps that ratio to a state-specific optimism setting, with a matching penalty in the value targets. Supporting contributions are a formal appendix (the coverage property, controller behavior, IQL as a special case), a benchmark comparison that keeps the losing rows, a memory analysis, and ablations.

6. **What is the research method?** The method is algorithm design with light formal analysis and benchmarking. UNIQ uses three heads per optimism level, and held-out transitions recalibrate the error scale every 5,000 training steps. The normalized uncertainty drives the per-state setting between 0.5 and 0.90/0.95. Two hand-set configurations cover nine simulated locomotion datasets over three seeds, with baselines copied from a 2023 benchmark report.

7. **What are the results?** UNIQ averages 85.2 across the nine tasks versus IQL's 81.6 and scores above the copied IQL figure on every task (by 0.1 to 8.1 points). It ranks third overall, behind two much heavier or more heavily tuned methods. Its measured memory use is 250 MB against an estimated 2.5 GB for the ensemble leader. The single-seed, four-task ablations temper these results. At matched settings (appendix table), the hand-set penalty coefficient appears to matter more than either calibration or adaptivity. In the main-text table, a single head per level, which gives no uncertainty signal, also nearly matches the full method (59.1 vs. 59.4).

8. **What are the impacts and limitations of the contributions?** UNIQ is a proof of concept that conformal calibration can be added cheaply to a standard learner, and it reports its losses explicitly. The conformal guarantee is marginal: it covers held-out Bellman residuals under one task-wide threshold, not individual states, and holds only at each recalibration of a moving ensemble. The gains are partly confounded with per-task penalty settings, the evidence is locomotion-only, and several numbers conflict across tables, text, and appendix.

9. **How were the results evaluated or validated?** The evaluation compares three-seed benchmark scores with eight baselines copied (not rerun) from a prior report. The ablations are single-seed with inconsistent labeling, and the speed and baseline memory estimates are analytic rather than measured. Realized coverage and the distribution of the uncertainty signal

are never reported, so the headline 3.6-point gain over IQL is plausible but weakly established, and the IQL comparison is not matched.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics discussion; the nearest thing is its scope section. Its language of distribution-free finite-sample guarantees could imply a safety assurance the method does not provide. The coverage concerns one-step Bellman residuals on held-out logged transitions, not the states the learned policy will visit. The controller also grows *more* optimistic wherever ensemble heads agree, which is dangerous if they share the same error. Data bias is inherited, validation is simulated, and no code release is mentioned. The small memory footprint could widen access.

11. **What themes are suggested by this research? (minimum 3)** *The conservatism–improvement trade-off*: the paper's premise is that one global setting is too careful in dense regions and may be too lax in sparse ones. *Uncertainty quantification and calibration*: the paper argues that raw disagreement needs a common scale. *Theory vs. practice*: conformal coverage on residuals is not shown to carry over to the derived signal or to states the policy visits.

Connection to the offline RL problem space and BCA: Of the papers reviewed here, UNIQ is closest to BCA: both bring conformal ideas to prediction-error-driven caution. UNIQ calibrates the errors but normalizes ensemble *spread* with the resulting quantile, so its guarantee does not formally extend to its uncertainty signal. BCA calibrates the errors against a fitted scale model with a periodically refreshed WBCP threshold, so width and signal describe the same quantity. Its guarantee is still marginal and on residuals, not per sample or on the policy.

Paper 9: Bayesian Flow Networks for Offline Trajectory Planning (BFN-RL)

BFN-RL swaps the diffusion model inside a trajectory planner for a Bayesian Flow Network, so that one kind of generator can plan over both grid-world (discrete) and robot-control (continuous) states.

1. **What is the citation?** Killingberg, L., & Langseth, H. (2026). *Bayesian flow networks for offline trajectory planning* (arXiv:2608.25163v1) [Preprint]. arXiv. `https://doi.org/10.48550/arXiv.2608.25163`

2. **What is the research area?** The area is offline RL treated as sequence generation rather than value learning. In the line of Diffuser and the Decision Diffuser, the planner generates a multi-step future state sequence conditioned on a desired return or goal. It then recovers actions with a learned inverse-dynamics model, a network that infers which action leads from one state to the next. With no critic, the inflated-value problem that drives the earlier papers does not arise in the same form.

3. **What is the research question?** Can a Bayesian Flow Network (BFN) serve as a single generator for offline trajectory planning that handles discrete and continuous states natively while staying competitive with diffusion-based planners in each setting?

4. **What is the significance of the question?** Many planning problems are discrete (grid navigation, key-and-door puzzles, box pushing) or mix discrete and continuous variables, yet diffusion planners are built for continuous vectors and need a separate categorical formulation for discrete states. One generator that covers both would simplify planner design, and a matched comparison can indicate whether the choice of generator deserves attention at all.

5. **What are the contributions reported in this paper?** The paper makes three contributions. First, BFN-RL is a planner whose generator refines the parameters of a distribution over a state trajectory instead of denoising a noisy one, so the same machinery applies to discrete and continuous spaces. Second, a discrete planner pairs a categorical BFN with a learned inverse-dynamics model, using an inpainting rule that pins known states during sampling. Third, a cross-domain study covers D4RL locomotion and five discrete tasks; on the discrete tasks, a categorical-diffusion baseline is matched on data, network, conditioning, protocol, and seeds. The paper claims no theory and mentions no code release.

6. **What is the research method?** This is an engineering study that designs and benchmarks a planner, with no theoretical analysis. The discrete testbed is three MiniGrid tasks, slippery FrozenLake, and two-box Sokoban, where the BFN and diffusion generators share data, network, conditioning, and seeds. For continuous control, a Gaussian BFN is scored on nine D4RL locomotion datasets against eight published baselines. Every experiment uses three seeds.

7. **What are the results?** In the matched discrete comparison, neither generator dominates. Four of the five tasks are close in success rate (e.g., 86.7% vs. 85.5% on FrozenLake), and BFN-RL wins clearly only on Sokoban (68.7% vs. 59.0%). On D4RL it averages 77.0, behind the Decision Diffuser (81.8), level with IQL (77), and just below CQL (77.6). It posts the best score in the table on halfcheetah-medium-expert (97.2) but falls below plain behavior cloning (pure imitation of the data) on walker2d-medium (66.9 vs. 75.3).

8. **What are the impacts and limitations of the contributions?** The paper is a proof of concept that BFNs can drive offline planners, with a reusable inpainting rule and a matched-baseline protocol designed to isolate the generator's effect. By the authors' own account, BFN-RL does not beat diffusion across the board. No component is ablated, and per-task settings such as temperature and step counts are reported without explanation of how they were chosen. The discrete datasets contain only successful trajectories, and in continuous control BFN-RL trails the Decision Diffuser by 4.8 points on average.

9. **How were the results evaluated or validated?** The discrete tasks use a matched protocol with three training seeds and 100 evaluation episodes per seed (200 on FrozenLake), reporting means and standard deviations. Continuous control uses D4RL normalized scores over three seeds, with every baseline number copied from the Decision Diffuser paper. The matched design is the strongest part. Confidence is limited by the three seeds, the lack of significance tests and ablations, and an undescribed hyperparameter selection.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics statement or limitations section. Its introduction motivates offline learning with self-driving cars and clinical treatment, where exploration is dangerous. Yet a generative planner can propose state sequences the system cannot reach, and conditioning on returns outside the data may push generation into unfamiliar regions with no uncertainty signal. Everything is simulated. Frequent replanning could act as a safeguard, but the paper presents it only as a planning detail.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: with no critic, the inflated-value problem does not arise in the same form; the planner samples fixed-horizon state sequences instead of learning values. *The conservatism–improvement trade-off*: the planner seeks improvement by conditioning its generator on a desired return or goal, yet on walker2d-medium its three-seed mean falls below the copied behavior-cloning score (66.9 vs. 75.3). *Theory vs. practice*: a matched baseline is designed to isolate the generator, but

copied baselines and three seeds weaken the continuous comparison, and per-task settings are unexplained and not ablated.

Connection to the offline RL problem space and BCA: BFN-RL represents the critic-free branch of the problem space. With no critic there are no Bellman residuals (the Bellman errors of Paper 5), so BCA's calibration cannot attach directly. One loose analogue is BFN-RL's fixed sampling temperature, which arguably plays the role of a global caution knob.

Paper 10: Conformal Prediction Beyond the Horizon: Distribution-Free Inference for Policy Evaluation (Conformal Policy Evaluation)

This paper adds a conformal calibration layer to a distributional RL model, so a policy's long-run return comes with a prediction interval checked against held-out data.

1. **What is the citation?** Gan, F., Lu, Y., Zhang, Y., & Liu, Y. (2025). Conformal prediction beyond the horizon: Distribution-free inference for policy evaluation. *Advances in Neural Information Processing Systems, 38*. Version reviewed: arXiv:2510.26026v1. `https://doi.org/10.48550/arXiv.2510.26026`

2. **What is the research area?** The paper applies statistics to RL. It uses conformal prediction (prediction intervals that assume no particular form for the data's distribution) for policy evaluation in infinite-horizon discounted problems, on-policy and off-policy (data from the evaluated policy or from a different one). It builds on distributional RL and connects to offline RL through off-policy evaluation from logged data.

3. **What is the research question?** How can prediction intervals for a target policy's infinite-horizon discounted return be built from finite logged trajectories to reach a chosen coverage level without depending on a correctly specified return model, both on-policy and off-policy?

4. **What is the significance of the question?** An expected return hides the spread of outcomes: a policy can look good on average and still do badly from some states. Distributional RL estimates the full return distribution, but its quantiles inherit function-approximation bias and can undercover (miss the true return more often than claimed) without warning. Conformal prediction offers model-agnostic validity, but its exchangeability assumption (that data order does not matter) fails for sequential data, and the infinite-horizon returns it needs as labels are never fully observed.

5. **What are the contributions reported in this paper?** The paper reports five contributions. It builds a pseudo-return label from k observed discounted rewards plus a tail sampled from the learned return distribution. It subsamples calibration transitions, replay-style, to weaken time dependence, and uses weighted resampling to correct for shifts in the state distribution and, off-policy, in action choice. It gives asymptotic coverage bounds that separate weight-estimation error from model error, and a simulation study with public code.

6. **What is the research method?** The paper combines algorithm design, asymptotic analysis, and Monte Carlo simulation. Quantile temporal-difference learning (a kernel density estimate on Mountain Car) fits the return distribution on training transitions, and each calibration transition is scored by the gap between its pseudo-return and the predicted value. The method is compared with the uncorrected quantile intervals on a two-state problem, a two-dimensional continuous problem, Mountain Car, and a 50-dimensional extension.

7. **What are the results?** In theory, coverage in the large-sample limit is at least the target minus a penalty: weight-estimation error plus a model-error term that shrinks geometrically in the rollout length k. In the two main simulations, conformal coverage stays near the 90% target (medians of roughly 0.89–0.92) while the raw quantile intervals undercover (roughly 0.84–0.87), though the conformal intervals are wider in the continuous example. On Mountain Car, on-policy coverage is near 0.90 but off-policy coverage stays near 0.85. In the two-state problem, coverage rises with k, from 0.87 at $k = 1$ to 0.94 at $k = 8$.

8. **What are the impacts and limitations of the contributions?** The paper shows that a conformal layer can sit on top of an ordinary distributional RL estimator using standard transition data. But the guarantees are asymptotic and hold on average over test states, and the authors call the choice of k underexplored. The score in the main experiments gives every state the same interval width. A second tuning parameter, which controls how intervals from repeated resamples are merged, was set using long simulated rollouts that are unavailable offline. The experiments are small and synthetic, and the one standard task falls short off-policy.

9. **How were the results evaluated or validated?** The theory rests on appendix proofs under Markov, stationarity, bounded-density, and overlap assumptions. The simulations use environments where long rollouts reveal the true return, so coverage is measured directly over 50–100 repetitions, against raw quantile intervals and one earlier conformal method. The design isolates coverage cleanly but is narrow, and at least one claim goes beyond the figures: the text calls Mountain Car coverage close to the target, while its off-policy medians sit near 0.85.

10. **What are the ethical implications? How are they addressed?** There is no ethics section, and the authors answer no to the NeurIPS checklist's broader-impact and safeguards questions, although the motivation is clinical treatment and autonomous systems. An interval labeled 90% can create unwarranted confidence when its assumptions fail. Because coverage is only an average over states, a patient subgroup can be undercovered while the overall number looks fine. The paper discusses neither risk, though its public code and its aim of exposing uncertainty are positives.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: weighted resampling is meant to correct for state shift and, off-policy, action shift, but off-policy coverage on Mountain Car stays near 0.85 against a 0.90 target. *Uncertainty quantification and calibration*: raw distributional quantiles undercover in the simulations, and held-out calibration narrows or closes the gap. *Theory vs. practice*: coverage survives an imperfect model only approximately and asymptotically, and the two tuning parameters that move empirical coverage were assessed against ground-truth returns that logged data cannot provide.

Connection to the offline RL problem space and BCA: This paper is an evaluation-side counterpart of BCA: both calibrate bootstrapped quantities (targets that lean on the model's own predictions) on held-out transitions. Its theory is a warning for BCA, since with one-step targets coverage of the true return is certified only up to the model's own error. Its main experiments use constant-width intervals, whereas BCA's fitted scale model lets the width vary per sample.

Paper 11: Bellman Calibration for V-Learning in Offline Reinforcement Learning (Bellman Calibration)

Bellman calibration checks whether a trained value function's numbers are systematically mis-scaled and, if so, adjusts them with a one-dimensional correction learned from logged transitions.

1. **What is the citation?** van der Laan, L., & Kallus, N. (2026). *Bellman calibration for V-learning in offline reinforcement learning* (arXiv:2512.23694v2) [Preprint]. arXiv. `https://doi.org/10.48550/arXiv.2512.23694`

2. **What is the research area?** The paper sits between statistics and RL. It studies post-hoc calibration of a trained value function for a fixed target policy, which is the policy-evaluation side of offline RL, and it carries classical calibration tools into the Bellman setting.

3. **What is the research question?** Can a trained value predictor be checked for systematic errors in its numerical scale and repaired by a single scalar transformation of its outputs, learned from logged transitions? The guarantees should be finite-sample and need neither Bellman completeness nor a correctly specified value class.

4. **What is the significance of the question?** Value estimates feed real decisions, so a scale that runs too high or too low can distort comparisons between policies even when the ranking of states is sensible. Existing guarantees for fitted value learning typically rest on strong structural conditions such as Bellman completeness, and the authors find much less model-agnostic theory for diagnosing and repairing a value function's scale after training.

5. **What are the contributions reported in this paper?** The paper reports four contributions. It defines Bellman calibration: states that share a predicted value should, on average, have a one-step backup (reward plus discounted next prediction) equal to that prediction. It gives a calibration-error measure with a doubly robust off-policy estimator (valid if either of two helper models is right), and a bound tying value error to calibration error plus what no rescaling can fix. It proposes Iterated Bellman Calibration, a post-hoc procedure in histogram, isotonic, and hybrid forms, and gives finite-sample bounds that assume neither completeness nor a correctly specified value class.

6. **What is the research method?** The paper is mainly theory and method, supported by simulations and a benchmark audit. The calibrator repeatedly regresses one-step backup targets on the original predictions, so every iterate stays a transformation of the initial value function. The simulated problems are built around specific failure modes. A frozen neural critic on the Deep OPE benchmark's Hopper data is then calibrated across eleven target policies and 20 random splits.

7. **What are the results?** In theory, calibration error is bounded, up to helper-model and iteration terms, by one-dimensional statistical rates whose leading term does not grow with the horizon. In simulation the corrections cut value error where the scale was wrong: by about a fifth under early stopping, by more in the mis-scaled-reward case, but by under a tenth with the saturating reward. On Hopper, the recommended hybrid calibrator cuts calibration error to about 0.78 of its raw level for the better-trained of two critics, though the final policy-value error barely moves (0.978 of its raw level). The linear and isotonic versions worsen that critic's Bellman diagnostics.

8. **What are the impacts and limitations of the contributions?** The paper offers a cheap, model-agnostic audit-and-repair step for any critic, and it explains when rescaling can help

(good ranking, wrong scale) and when it cannot. The authors stress that calibration is not accuracy, and the fully iterated isotonic version lacks a value-error guarantee. The simulations use only clipped importance weights, a special case of the doubly robust targets the theory centers on. On the one standard benchmark the policy-value error barely improves.

9. **How were the results evaluated or validated?** The results are validated by appendix proofs, by simulations scored against Monte Carlo ground truth over 100 replications, and by a Hopper audit with shared splits, held-out scoring, and bootstrap intervals. The protocol guards against leakage and reports cases where calibrators hurt. But the simulated failure modes are the ones the method is designed to fix, and several settings go unreported. The mis-scaled-reward results differ without explanation between a figure and a table (0.42 vs. 0.73–0.75), and no external recalibration or evaluation method is compared.

10. **What are the ethical implications? How are they addressed?** None are discussed. Because calibration conditions only on the predicted value, errors for one group can cancel against opposite errors elsewhere at the same prediction level, so a low calibration error could hide systematic harm to a subpopulation. A small calibration error might also be over-read as validation. The paper is frank about technical risks but silent on subgroup effects and deployment safeguards, and all empirical evidence comes from simulators.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: a target policy's Bellman backups are estimated from another policy's data, with doubly robust targets in theory and clipped importance weights when the simulations fit the calibrator. *Uncertainty quantification and calibration*: calibration is carried over to long-horizon value predictions through a post-hoc, one-dimensional fix that leaves the critic untouched; it concerns the reliability of the mean rather than interval coverage. *Theory vs. practice*: on the better-trained Hopper critic, the calibrators with value-error guarantees cut calibration error while those without raised it, though elsewhere both kinds helped.

Connection to the offline RL problem space and BCA: Bellman calibration complements BCA: BCA reads uncertainty from the size of Bellman residuals, while this paper targets their average bias among states with similar predicted values. Its calibration error could serve as a companion diagnostic for BCA's host critics, though its guarantees cover a frozen critic and a fixed policy, not actor-critic training.

Paper 12: Weighted Bayesian Conformal Prediction (WBCP)

WBCP gives a conformal threshold its own error bar: alongside each prediction interval it reports how firmly the calibration data support that interval's width.

1. **What is the citation?** Lou, X., & Luo, P. (2026). *Weighted Bayesian conformal prediction* (arXiv:2604.06464v2) [Preprint]. arXiv. `https://doi.org/10.48550/arXiv.2604.06464`

2. **What is the research area?** The area is statistical machine learning rather than RL. Within conformal prediction, the paper sits where two approaches meet. Weighted conformal methods reweight calibration scores under covariate shift (test inputs distributed differently from the calibration inputs). Bayesian quadrature conformal prediction (BQ-CP, Paper 18) places a posterior over the threshold.

3. **What is the research question?** Can the posterior over the conformal threshold that BQ-CP

provides for i.i.d. data be extended to weighted conformal prediction? The aim is one procedure that stays valid under covariate shift or local weighting and also reports how reliable its own threshold is.

4. **What is the significance of the question?** A conformal threshold is a single number that says nothing about how firmly the calibration data pin it down. That matters wherever interval width drives decisions and calibration data are unevenly spread. The authors argue that the two existing remedies had not been combined: BQ-CP gave a threshold posterior but needed exchangeable data, while weighted conformal prediction handled shift but returned only a point estimate.

5. **What are the contributions reported in this paper?** The paper reports four contributions. WBCP replaces BQ-CP's flat Dirichlet model over the gaps between sorted calibration scores with a weighted one scaled by Kish's effective sample size (how many equally weighted points the weighted sample is worth). It returns a conservative threshold together with a posterior spread. Four theoretical results, given as proof sketches, address why the effective sample size is the variance-matching scaling and when BQ-CP's guarantees carry over. The paper also gives a spatial version with an adaptive bandwidth (AdaGeoBCP) and an empirical study on a synthetic field and 13 geospatial datasets.

6. **What is the research method?** The paper combines algorithm design and asymptotic proof sketches with benchmarking and ablations. Each of many Monte Carlo draws samples from the weighted Dirichlet and reads off a threshold. A high quantile of those thresholds sets the interval, and their spread gives the reliability measure. Tests target 90% coverage against standard, weighted, and Bayesian conformal baselines.

7. **What are the results?** Across the 13 real datasets, the spatial version of WBCP reaches a mean coverage of 0.925, essentially tied with BQ-CP (0.928) and above standard conformal prediction (0.903), while the frequentist spatial baseline undercovers (0.824). Its distinguishing output is a reliability measure that varies by location, which BQ-CP cannot give. The costs are visible on the Seattle house-price data, where WBCP has the widest intervals (24.9 vs. 20.4 for BQ-CP) and its coverage depends on the chosen posterior level (0.873 at $\beta = 0.5$, 0.95 at $\beta = 0.9$).

8. **What are the impacts and limitations of the contributions?** WBCP turns a conformal threshold into a distribution under any weighting scheme, so each interval comes with a cheap measure of how well the data support it. The authors state three limitations. Correlated residuals inflate the effective sample size, misspecified weights may void the coverage result, and the theorems are sketches under an assumed model that need a large effective sample size. Yet the method ran with roughly 10–13 effective points on some data, and the random splits test no genuine covariate shift.

9. **How were the results evaluated or validated?** The evaluation uses a synthetic field with known noise, a sweep over spatial correlation lengths, 13 real datasets with 10 random splits each, a 50-split study on one dataset, and an ablation. The metrics are coverage, interval width, and two reliability diagnostics. The breadth and the clean ablation are strengths. The weaknesses are no significance tests, a lenient rule for counting a dataset as meeting the target, no check of local coverage, and one figure whose values conflict with the matching table.

10. **What are the ethical implications? How are they addressed?** There is no ethics statement. The applications tested (house prices, disease prevalence, vote shares) can shape

lending, health-resource allocation, or political targeting. Because sparsely sampled areas are flagged as less reliable, the reliability map could guide new data collection, or could be used to justify withholding services from places already poorly measured. The outputs also look like rigorous certificates although they rest on an assumed model and correct weights. The authors soften their claims but do not discuss misuse or equity, and no code is linked.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: calibration scores are reweighted for test inputs that differ from the calibration inputs, though the random splits test no genuine covariate shift. *Uncertainty quantification and calibration*: each threshold reports how well the calibration data support it, and the threshold posterior is reported to widen where calibration support is thin. *Theory vs. practice*: beyond the exchangeable case, WBCP's guarantees rest on an assumed weighted Dirichlet model and a large effective sample size.

Connection to the offline RL problem space and BCA: WBCP is the calibration procedure BCA runs, although BCA's code follows the paper's later v3 revision rather than the v2 reviewed here. BCA uses it with uniform weights, which makes it essentially BQ-CP, so the calibrated width inherits an exchangeability assumption that policy-induced shift and correlated transitions can break. With uniform weights WBCP also yields one global threshold, so BCA's per-sample adaptivity comes from its fitted scale model, not from WBCP.

Paper 13: Off-Policy Deep Reinforcement Learning without Exploration (BCQ)

BCQ shows that standard deep RL algorithms can fail badly when learning from a fixed dataset, and counters this by letting the policy choose only among actions that resemble the recorded ones.

1. **What is the citation?** Fujimoto, S., Meger, D., & Precup, D. (2019). Off-policy deep reinforcement learning without exploration. In *Proceedings of the 36th International Conference on Machine Learning* (Proceedings of Machine Learning Research, Vol. 97, pp. 2052–2062). PMLR. Version reviewed: arXiv:1812.02900v3. `https://doi.org/10.48550/arXiv.1812.02900`

2. **What is the research area?** The area is offline RL, which the paper calls batch RL (learning from a fixed dataset with no further interaction), for continuous control. BCQ is an early policy-constraint method: it restricts the policy to actions close to those in the data. It sits between imitation learning (copying the data) and off-policy value learning.

3. **What is the research question?** Why do off-policy deep RL algorithms such as DDPG and DQN, which in principle can learn from data gathered by any policy, fail when trained on a fixed dataset they did not collect themselves? Can restricting the policy to actions similar to the recorded ones make learning from such data work in continuous control?

4. **What is the significance of the question?** RL needs to learn from a fixed dataset wherever collecting data is costly, risky, or slow. Imitation learning works when the demonstrations are good but tends to fail on suboptimal ones. Off-policy algorithms seemed to fill that gap, but in practice they train on replay data closely tied to their own current policy, which hides how they behave on data they did not collect.

5. **What are the contributions reported in this paper?** The paper makes four contributions. It diagnoses *extrapolation error*: the critic assigns unrealistic values to state-action pairs the dataset lacks, and without new data these are never corrected. It gives a theory for finite (tabular) problems: in a deterministic environment a policy confined to recorded state-action

pairs can be evaluated without this error, and a constrained Q-learning (BCQL) converges to the best such policy. It presents BCQ as the first deep continuous-control method that learns from arbitrary fixed data. It also provides four fixed-data test settings (final buffer, concurrent, imitation, imperfect demonstrations) with public code.

6. **What is the research method?** The paper combines diagnostic experiments, tabular proofs, and benchmarking. BCQ fits a generative model (a conditional variational auto-encoder) to the recorded actions, samples $n = 10$ candidates per state, lets a perturbation network shift each by at most Φ, and picks the one the critic scores highest. The tests use three simulated locomotion tasks (HalfCheetah, Hopper, Walker2d) and four datasets from DDPG agents: a full replay buffer, a buffer shared with the collecting agent, expert data, and noisy expert data. The baselines are DDPG, a discretized DQN, and two behavior-cloning (pure imitation) variants.

7. **What are the results?** DDPG trained on fixed data does markedly worse than the agent that collected it, even when both learn from the same replay buffer. Its value estimates are unstable or diverge, while BCQ's stay stable. Read off the learning curves, BCQ's return far exceeds the data's average on noisy demonstrations (about 2,300 vs. 450 on Hopper, 6,100 vs. 2,550 on HalfCheetah) and also exceeds it on final-buffer data (about 8,300 vs. 6,000 on HalfCheetah). On clean expert data behavior cloning is best on two of three tasks, where BCQ ends slightly below the expert. On two toy tasks with random data, plain DDPG also learns.

8. **What are the impacts and limitations of the contributions?** The paper names a failure of offline value learning that it calls fundamental, and shows one algorithm with one set of hyperparameters both imitating expert data and improving on noisy data. The survey in Literature Review 1 cites it for the extrapolation-error observation and as a policy-constraint method. However, the key guarantees cover tabular, deterministic problems, not the deep algorithm, and the authors report instability on one task under slightly different hyperparameters. On expert data, returns fall sharply as Φ grows, yet Φ is hand-set and, like n, is the same in every state whatever the data coverage. The main benchmark data all come from DDPG agents in simulation.

9. **How were the results evaluated or validated?** The experiments use five seeds, averaging 10 noise-free evaluation episodes every 5,000 steps, and the learning curves carry half-standard-deviation bands. Value estimates are compared with Monte Carlo returns, and appendix studies vary Φ and try critic-ensemble uncertainty in place of the generative model. There are no result tables or significance tests, and no earlier batch RL method is among the baselines. The Φ used in the main runs is not stated in the text, and the appendix compares Φ settings by simulator returns, which a purely offline user would lack.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics or broader-impact discussion. The nearest it comes is a remark that fixed-data learning suits settings where data collection is costly or risky. A policy held close to logged behavior may carry over biases and unsafe habits from the log. The guarantee of at least matching the behavior policy is proved only for tabular, deterministic problems, so it does not cover the deep algorithm. All evidence is simulated. Public code and the authors' report of instability help, but deployment risks are not discussed.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: values for actions the data do not contain can be badly wrong and cannot be corrected offline (the paper's extrapolation error). *The conservatism–improvement trade-off*: the policy is kept close

to recorded actions, and Φ and n slide the method between pure imitation and unconstrained Q-learning. *Theory vs. practice*: the key guarantees are proved for tabular, deterministic problems, not for the deep algorithm that is tested.

Connection to the offline RL problem space and BCA: BCQ is an early form of the "stay near the data" policy-constraint idea that two of BCA's hosts, TD3+BC and ReBRAC, carry on as a plain penalty. Its constraint (Φ, n) has one strength for every state; BCA keeps a host's global coefficient but modulates the regularization per sample with a calibrated width. In BCQ's appendix, uncertainty from ensembles of 4 or 10 critics did not by itself keep the policy on expert data (one Hopper test), a caution for any method that leans on an uncertainty signal, BCA included.

Paper 14: Stabilizing Off-Policy Q-Learning via Bootstrapping Error Reduction (BEAR)

BEAR traces much of Q-learning's failure on fixed datasets to value targets at actions the data lack, and keeps the policy approximately within the actions the data support without forcing it to copy their frequencies.

1. **What is the citation?** Kumar, A., Fu, J., Soh, M., Tucker, G., & Levine, S. (2019). Stabilizing off-policy Q-learning via bootstrapping error reduction. *Advances in Neural Information Processing Systems, 32*. Version reviewed: arXiv:1906.00949v2. `https://doi.org/10.48550/arXiv.1906.00949`

2. **What is the research area?** The area is offline RL for continuous control. BEAR belongs to the policy-constraint family with BCQ (Paper 13): it restricts which actions the policy and the value targets may use, where CQL (Paper 1) instead lowers the critic's scores. Its distinguishing position is support matching (stay within the set of recorded actions) rather than distribution matching (also copy how often each was taken).

3. **What is the research question?** Why do standard Q-learning and actor-critic methods become unstable on a fixed dataset, even a large or expert one? Can restricting the actions used in value updates to the data's support, instead of to the behavior policy's exact distribution, remove that instability while still letting the policy improve on the data?

4. **What is the significance of the question?** Logged data are plentiful in areas such as driving and recommendation, where fresh trial and error is costly. Yet off-policy methods of the time could fail on a fixed dataset even when an expert collected it. The failure does not behave like ordinary overfitting: in the one test shown (HalfCheetah), growing the dataset from a thousand to a million samples does not cure it. BCQ, an earlier fix, is reported to stay so near the recorded behavior that it gains little over imitation on the mediocre data the authors consider the common practical case.

5. **What are the contributions reported in this paper?** The paper makes four contributions. It diagnoses "bootstrapping error," which enters when a value target is computed at an out-of-distribution action (one the data do not cover) and then accumulates through repeated backups. It analyzes a backup restricted to a chosen set of policies, giving a bound that splits the final error into a distribution-shift term and a suboptimality term. It presents BEAR, which keeps the actor approximately within the data's support by limiting the maximum mean discrepancy (MMD, a sample-based distance between distributions). It also compares methods across random, medium-quality, and expert data, with ablations.

6. **What is the research method?** The paper combines theory, algorithm design, and benchmarking. The theory adapts known error-propagation bounds from approximate dynamic programming and is illustrated on a small gridworld. The actor maximizes the lowest of several critics' scores subject to an MMD limit ($\varepsilon = 0.05$), enforced by a self-adjusting penalty weight (a Lagrange multiplier). Experiments cover five MuJoCo tasks with one-million-transition datasets from a uniformly random policy, a partially trained SAC policy, or a converged one (Humanoid has random and medium data only). Baselines are naive off-policy RL (TD3), BCQ, KL-control, a static variant of DQfD, and behavior cloning.

7. **What are the results?** On medium-quality data BEAR ends ahead on Walker2d (about 1,800 vs. 800 for BCQ), Hopper, and Humanoid, narrowly ahead on HalfCheetah, and near 700 on Ant, just under the dataset's average return. There, a KL constraint in place of MMD collapses on Hopper and Walker2d and trails MMD even when heavily tuned. On random data BEAR beats the dataset average on all five tasks, yet naive RL scores higher on HalfCheetah (about 3,300 vs. 2,900) and Hopper. On expert data BEAR, BCQ, and behavior cloning are roughly level. All numbers are approximate readings of learning curves.

8. **What are the impacts and limitations of the contributions?** The paper names the instability, offers a mechanism for it, and separates matching the data's support from matching its distribution. CQL (Paper 1) later uses BEAR as a baseline. The authors note that performance can still decay over long runs with no stopping rule and that action constraints can be too conservative, especially on mixed-policy data. They also note that support matching in high-dimensional action spaces still lacks a theoretically robust method. Beyond these, the bound assumes policies that truly stay in the support, which a sampled, state-averaged MMD only approximates, and one threshold serves all states however well each is covered.

9. **How were the results evaluated or validated?** The theory is proved in an appendix. The experiments report learning curves averaged over five seeds, with shaded bands for spread, and all methods share the same datasets. There are no tables or significance tests. The paper's claim to match or exceed the state of the art in all cases is not borne out on random data, where naive RL averages higher on two tasks. Ensemble size and other settings are not reported, and the first term of Theorem 4.1's bound is half what its proof derives.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics or broader-impact discussion. The nearest content is its motivation (driving, recommendation, robotics) and a hope of training on static data before real-world deployment. A policy confined to the data's support can inherit biases of the logged behavior, and the bound concerns value error, not safe behavior. Because performance can decay late in training with no offline stopping rule, a policy's quality may be hard to confirm without live trials. All evidence is simulated. The authors report the decay, but bias, safeguards, and misuse are not addressed.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: value targets at unseen actions cannot be corrected offline, and backups spread the error. *The conservatism–improvement trade-off*: a wider allowed policy set raises the bound's shift term and lowers its suboptimality term. *Theory vs. practice*: the bound assumes policies that truly stay in the support, which a sampled, state-averaged MMD only approximates.

Connection to the offline RL problem space and BCA: BEAR formalizes, for action constraints, a trade-off that also bears on caution mechanisms such as BCA's. Admitting too many actions lets errors from unsupported ones spread through backups, and admitting too few leaves the policy

little room to improve. BEAR sets this balance with one threshold and one learned penalty weight, enforced on average over states, whereas BCA varies each host's regularization per sample by a calibrated width while keeping the host's global coefficient. BEAR is not one of BCA's four hosts.

Paper 15: Addressing Extrapolation Error in Deep Offline Reinforcement Learning (BRr)

BRr learns the value of the policy that recorded the data, so its training targets use only recorded actions; its one improvement step comes at deployment, with a ranking penalty meant to keep unseen actions from looking attractive.

1. **What is the citation?** Gulcehre, C., Gómez Colmenarejo, S., Wang, Z., Sygnowski, J., Paine, T., Żołna, K., Chen, Y., Hoffman, M., Pascanu, R., & de Freitas, N. (2020). Addressing extrapolation error in deep offline reinforcement learning. In *Offline Reinforcement Learning Workshop at NeurIPS 2020*. Version reviewed: the workshop paper. `https://offline-rl-neurips.github.io/pdf/41.pdf` Also available at `https://openreview.net/forum?id=0CRKCul3eKN`; a later version appears as *Regularized behavior value estimation* (arXiv:2103.09575).

2. **What is the research area?** The area is offline RL for problems with a finite set of actions. The paper targets extrapolation error (BCQ's term, Paper 13): the mistakes a neural critic makes when scoring state-action pairs the data barely cover. Its remedy evaluates only the behavior policy (the policy that recorded the data), which resembles the in-sample idea of Paper 3. It also pushes down the scores of unseen actions, which the paper groups with BCQ and CQL as behavior constraining.

3. **What is the research question?** Can offline value learning avoid runaway overestimation on poorly covered state-action pairs by estimating the value of the behavior policy instead of the optimal one and taking a single greedy improvement step only at deployment? With suitable regularization, does this match or beat existing offline methods as data coverage varies?

4. **What is the significance of the question?** Offline RL is aimed at settings such as healthcare and self-driving cars, where repeated interaction is costly or unsafe while logs are plentiful. But the max over actions in Q-learning includes actions the data lack, and the critic's guesses for them are sometimes too high. Bootstrapping (building targets from the critic's own predictions) spreads the error, and with no new data nothing corrects it. The paper adds that Double DQN, a standard fix for overestimation, targets a different cause and does not help here.

5. **What are the contributions reported in this paper?** The paper contributes three techniques. Behavior value estimation (B) is a SARSA-style update whose target uses the next action actually recorded, followed by a greedy policy at test time. The ranking regularizer (R) is a hinge penalty asking the recorded action to outscore every other action, weighted toward high-return trajectories. The reparametrized critic (r) is a bounded output times one learned scale, penalized when values exceed observed returns. The paper also provides new low-coverage bsuite and DeepMind Lab datasets (release promised) and, as its main formal result, a hand-built four-state example in which offline Q-learning diverges. The later arXiv version names the approach R-BVE.

6. **What is the research method?** The paper combines algorithm design with benchmarking. Of its three testbeds, the bsuite tasks have random actions injected at five noise levels and only 10% of the data kept. On RL Unplugged Atari, methods are tuned on one set of games and

reported on another. DeepMind Lab supplies four partially observable levels, plus one level recorded at four noise levels. Baselines include DDQN, BCQ, REM, CQL, IQN, offline R2D2, and behavior cloning (BC, pure imitation). Ablations vary the backup (Q-learning or B), toggle each regularizer, and shrink the Atari data to 10% and 1%.

7. **What are the results?** On the held-out Atari games QRr, Q-learning with both regularizers, scores highest (median normalized score 108.2; IQN 104.8, CQL 98.9) and also edges BRr on the tuning games (113.4 vs. 110.7). Behavior value estimation helps when coverage is thin: on 10% of the Atari tuning-game data B scores 58.6 against DDQN's 14.0, though DDQN is slightly ahead on the full data (94.1 vs. 88.2). BRr leads at all five cartpole noise levels (933.8 vs. 383.7 for the best baseline at zero noise), and BR's mean return is above CQL's and offline R2D2's on all four DeepMind Lab levels. BR loses to BC on data recorded with little or no noise (17.1 vs. 28.0 with none).

8. **What are the impacts and limitations of the contributions?** The paper gives evidence that one greedy step on a behavior-policy critic can compete with dedicated offline methods when coverage is thin. The authors note that one step can be too conservative in data-rich settings. The headline Atari number comes from QRr, which reinstates the max operator, and no BRr score is given for the held-out games. The featured variant changes by domain (QRr on Atari, BR on DeepMind Lab after preliminary live results), and all tests use discrete actions. The greedy step still relies on scores of untested actions, guarded by a heuristic penalty with no stated guarantee. On each single-level noise dataset BR's return (17.1–31.8) is below that dataset's average return (29.7–40.4), so improvement over the behavior policy is not shown there.

9. **How were the results evaluated or validated?** The stated protocol sweeps 5–10 hyperparameter settings per method, trains on five datasets from five behavior seeds, evaluates in the real environment, and reports each method's best setting. Atari scores are medians with bootstrapped 25–75 percentile bars, and DeepMind Lab results carry standard deviations. There are no significance tests, and spreads sometimes overlap (BR 13.4 ± 11.8 vs. CQL 9.3 ± 2.5). Best settings are picked by live evaluation (per level on DeepMind Lab), and most Atari baselines are taken from the benchmark paper. Figure 7 and Table 8 disagree (swapped level labels, and an untabulated BRr bar above the BR bar the caption calls best).

10. **What are the ethical implications? How are they addressed?** The paper has no ethics or broader-impact section and only motivates offline RL with healthcare and self-driving cars, where live trial and error is costly, unsafe, or unethical. The deployed policy is greedy over scores that include actions the data never tested, with a heuristic penalty as the safeguard. The ranking term favors whatever was done in high-return episodes, so biases in those logs could be reinforced. Settings were chosen by live evaluation, which the motivating domains do not allow. The paper treats test-time extrapolation as a technical problem but does not discuss bias or deployment safeguards. All evidence is simulated.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: the behavior-value variants keep it out of the training target by design, because the target bootstraps only from recorded actions and improvement is left to one greedy step at deployment. *The conservatism–improvement trade-off*: one step scored higher in the low-data tests but can be too cautious when data are rich, where regularized Q-learning edged ahead on Atari. *Theory vs. practice*: the motivating domains rule out live trials, yet the best settings were picked by live evaluation, and the ranking of methods flips with coverage, noise, and data size.

Connection to the offline RL problem space and BCA: This workshop paper predates IQL (Paper 3, one of BCA's hosts) and shares its in-sample principle: the bootstrap target is built only from recorded actions, though the ranking penalty does score unseen actions to push them down. It differs by improving in a single step at deployment. Its penalty has one global coefficient, weighted per sample only by trajectory return (data quality, not critic uncertainty), whereas BCA draws its per-sample signal from the size of Bellman residuals and keeps each host's global coefficient.

Paper 16: Uncertainty Weighted Actor-Critic for Offline Reinforcement Learning (UWAC)

UWAC estimates the critic's uncertainty about each training sample and gives uncertain samples less weight in the critic and actor losses, so that value guesses about situations the data do not cover destabilize training less.

1. **What is the citation?** Wu, Y., Zhai, S., Srivastava, N., Susskind, J., Zhang, J., Salakhutdinov, R., & Goh, H. (2021). Uncertainty weighted actor-critic for offline reinforcement learning. In *Proceedings of the 38th International Conference on Machine Learning* (Proceedings of Machine Learning Research, Vol. 139, pp. 11319–11328). PMLR. Version reviewed: arXiv:2105.08140v1. `https://doi.org/10.48550/arXiv.2105.08140`

2. **What is the research area?** The area is uncertainty-based offline RL: actor-critic methods that estimate how unreliable the critic is and act on that estimate. UWAC is model-free (it learns no model of the environment) and builds on BEAR (Paper 14), a policy-constraint method that aims to keep the policy's actions within the data's support. Its uncertainty tool is Monte Carlo dropout, from Bayesian deep learning, which leaves dropout's random masking on at prediction time and measures how much repeated predictions disagree.

3. **What is the research question?** Can a critic's own dropout-based uncertainty identify out-of-distribution (OOD) state-action pairs, meaning ones the dataset does not cover? And does down-weighting those pairs in the critic and actor losses stabilize offline training and raise returns without adding new loss terms or a learned environment model?

4. **What is the significance of the question?** When training bootstraps (builds value targets from the critic's own predictions) from an action or state the data do not cover, the value there can be arbitrary. The error can then spread to other states through the Bellman backup. The authors argue that the usual remedy, strong constraints or penalties that hold the policy to the data, avoids unseen actions even when the critic could score them accurately. They say this often leaves policies pessimistic and suboptimal.

5. **What are the contributions reported in this paper?** The main contribution is a reweighting that makes uncertain samples count for less. Each critic update is scaled by the inverse of the estimated variance of its target value, and the actor's objective by the inverse variance of the critic's value for the proposed action. The paper also gives a low-overhead variance estimate from Monte Carlo dropout, an appendix convergence argument, evidence that dropout variance flags OOD samples, and a proposed explanation of why the BEAR baseline's values diverge. The reviewed version links the baseline's code but not UWAC's.

6. **What is the research method?** The method is algorithm design with a short convergence argument, then experiments. UWAC adds dropout to BEAR's critics and takes the variance across 100 stochastic forward passes of the target critics as its uncertainty. Each sample's weight

is β times the inverse variance (clipped at 1.5). BEAR's settings are otherwise kept, with spectral normalization (a weight constraint the authors use to keep the critic's values bounded) added on Adroit. Tests cover LunarLander data with a region of states deleted, D4RL locomotion and Adroit hand-manipulation datasets, and ablations.

7. **What are the results?** Dropout uncertainty is higher in LunarLander's deleted regions and separates recorded from random actions on walker2d-expert with an AUC of 0.845 (0.5 is chance). On Adroit, UWAC has the best mean on 8 of 12 datasets and improves on BEAR on human demonstrations (pen-human 65.0 vs. -1.0), while BEAR's value targets climb to the order of 10^{12}. On locomotion it leads on 6 of 15 datasets, mostly medium-expert and expert. It trails model-based methods on medium-replay data (hopper 39.4 vs. MOReL's 93.6) and falls below BEAR on halfcheetah-random (14.5 vs. 25.1).

8. **What are the impacts and limitations of the contributions?** UWAC shows that per-sample uncertainty weighting can be added to an existing actor-critic cheaply (reportedly under 1.5 times BEAR's training time). Together with spectral normalization, which alone also bounds the critic in the ablation, it keeps value targets from diverging on narrow Adroit data. But the dropout variance is never calibrated and is scaled by a swept β and a clip. As the authors note, the convergence argument covers inverse standard deviation, not the clipped inverse variance actually used. Only one host algorithm is benchmarked. Several human and cloned scores stay near zero, and UWAC's pen-human return itself decays from about 1,700 to about 700 during training.

9. **How were the results evaluated or validated?** The results are five-seed D4RL means and standard deviations, with training curves against BEAR. Adroit ablations include spectral normalization alone and dropout without weighting, which does sharply worse on most tasks. Baseline scores appear quoted from published tables, mostly without spreads, and evaluation episodes, dropout rate, and the scored checkpoint are unstated. BEAR is listed at a normalized 98.6 on relocate-expert although Figure 1 shows its average return collapsing to near zero. The text claims state-of-the-art results on every non-human Adroit dataset, yet other methods lead on three cloned ones.

10. **What are the ethical implications? How are they addressed?** The paper has no ethics or broader-impact statement and no limitations section, and all its evidence is simulated. Down-weighting poorly covered samples lets well-covered behavior shape the policy, so biases in the log could carry over. The uncertainty score could also be read as a safety signal although it is uncalibrated and imperfect by the paper's own test (AUC 0.845), so some unsupported samples may keep high weight. None of this is discussed, and more stable training is at most an indirect benefit.

11. **What themes are suggested by this research? (minimum 3)** *Distribution shift*: value targets built on unsupported actions are blamed for BEAR's diverging critic on narrow data. *The conservatism–improvement trade-off*: each sample's weight shrinks with its own estimated uncertainty, and this weighting is added on top of BEAR's support constraint. *Uncertainty quantification and calibration*: dropout variance is checked as an OOD detector (heat maps, one AUC) but never calibrated or compared with actual value error.

Connection to the offline RL problem space and BCA: UWAC is an earlier instance of BCA's central move: modulating an existing algorithm's training objective per sample by an uncertainty signal. UWAC scales the critic and actor losses of one host (BEAR) by an inverse Monte Carlo dropout

variance that is never calibrated. BCA instead derives a conformally calibrated width from Bellman residuals and uses it to modulate one loss term in each of four hosts. Unlike BCA's current design, UWAC has completed D4RL results.

Paper 17: Uncertainty-Based Offline Reinforcement Learning with Diversified Q-Ensemble (EDAC)

This paper shows that SAC, an online algorithm, can outscore the tested offline baselines on D4RL locomotion tasks by trusting only the lowest of many critics, and that with a diversity term (EDAC) tens of critics can replace hundreds on some datasets.

1. **What is the citation?** An, G., Moon, S., Kim, J.-H., & Song, H. O. (2021). Uncertainty-based offline reinforcement learning with diversified Q-ensemble. *Advances in Neural Information Processing Systems, 34*, 7436–7447. Version reviewed: arXiv:2110.01548v2. `https://doi.org/10.48550/arXiv.2110.01548`

2. **What is the research area?** The area is model-free, uncertainty-based offline RL for continuous control. Caution comes from disagreement among an ensemble of critics (several value networks trained side by side) instead of tying the policy to the recorded behavior (BCQ, BEAR, BRAC) or marking down unseen actions directly (CQL). The paper builds on soft actor-critic (SAC), a standard online algorithm, and on clipped Q-learning, the online habit of trusting the lower of two critics.

3. **What is the research question?** Can disagreement within an ensemble of critics, applied through clipped Q-learning's take-the-minimum rule, supply on its own the pessimism offline RL needs, with no model of the behavior policy and no sampling of unseen actions? And can the number of critics this requires be cut to a practical size?

4. **What is the significance of the question?** The authors argue that earlier methods typically need an accurate estimate of the behavior policy (the one that collected the data) or samples of out-of-distribution (OOD) actions, meaning actions the data do not contain. Each can be a non-trivial problem in itself. They add that those methods discourage unseen actions wholesale, whether or not the critic can score them reliably, which can hold the policy too close to mediocre data. A penalty tied to the critics' own confidence could be selective, but the plain ensemble version can demand hundreds of networks.

5. **What are the contributions reported in this paper?** The paper makes three contributions. First, SAC-N is SAC with N critics instead of two, all trained toward a target built from the lowest of their predictions. The paper gives evidence that this minimum acts like a lower confidence bound (the ensemble average minus a multiple of its spread). Second, it proposes an explanation of why some datasets need hundreds of critics: when the critics' slopes with respect to the action line up, nearby unseen actions can escape the penalty. Third, EDAC (Ensemble-Diversified Actor Critic) adds a term, weighted by η, that pushes those slopes apart at dataset actions. The reviewed version links no code of its own.

6. **What is the research method?** The method is algorithm design with a short supporting analysis under simplifying assumptions, then benchmarking. The tests cover 18 D4RL Gym locomotion datasets and eight Adroit robotic-hand datasets. The baselines include behavior cloning (BC, pure imitation), plain SAC, REM (an earlier critic-ensemble method), and CQL.

Diagnostics compare the penalty on dataset vs. random actions and find the smallest workable ensemble per dataset.

7. **What are the results?** On the 18 Gym datasets EDAC averages 85.2 and SAC-N 84.5, against 73.7 for the rerun CQL, with the largest gains on the medium datasets (hopper-medium: 101.6 vs. 61.9). On Gym the two score alike, and EDAC's gain is in cost: it uses 50 critics instead of 500 on hopper-medium, where 50 undiversified critics score 7.3. CQL wins halfcheetah-random (31.3 vs. 28.4), and EDAC trails SAC-N on hopper-random and walker2d-random. On Adroit, EDAC does well on the pen tasks (pen-cloned: 68.2 vs. CQL's published 40.3) but stays near 10 on door and below 1 on hammer and relocate.

8. **What are the impacts and limitations of the contributions?** The paper shows that on D4RL Gym tasks critic disagreement alone can supply enough pessimism to outscore CQL, and that diversity can stand in for ensemble size on some datasets. The authors report that SAC-N can need 500 critics and that a large η can hurt on suboptimal data. Beyond that, N and η are still chosen per environment or dataset (the chosen η runs from 0, i.e., off, to 1,000). The tenfold saving holds on only a few datasets, with none on halfcheetah. The spread is not checked against actual value errors, the analysis gives no performance guarantee, and no maze or kitchen tasks are tested.

9. **How were the results evaluated or validated?** The results are D4RL normalized scores averaged over four seeds, with a $\pm$ spread the paper does not define and no stated number of test episodes. CQL is rerun from official code beside its published scores. EDAC's Gym settings come from small grids (three values each for N and η) by an unstated criterion, presumably test returns, which a true offline user lacks. There are no significance tests, the Adroit stopping point (200,000 steps) is read off evaluation curves, and the rerun CQL is far below its published score on some tasks (hopper-medium: 61.9 vs. 86.6).

10. **What are the ethical implications? How are they addressed?** The reviewed version has no ethics, broader-impact, or limitations section; its introduction only cites autonomous driving and healthcare as settings where exploration can do damage. The method lets the policy leave the data wherever the critics agree, which is arguably a risk. Critics trained on the same data can share an error, so agreement does not show the value is right. Settings also appear to be tuned on test rollouts, and all evidence is simulated. Pessimism is an indirect step toward safer learning, but none of these risks is discussed.

11. **What themes are suggested by this research? (minimum 3)** *The conservatism–improvement trade-off*: the penalty grows with disagreement at each state-action pair instead of treating every unseen action alike. *Uncertainty quantification and calibration*: critic disagreement is the uncertainty signal, and its scale is not checked against real errors. *Theory vs. practice*: the reported runs get their pessimism from tens to hundreds of critics, and N and η are still chosen per environment or dataset by an unstated criterion.

Connection to the offline RL problem space and BCA: EDAC is not a BCA host but is among the baselines in the Unifloral codebase bundled with BCA's repository. It represents the ensemble route to per-sample caution, the large-ensemble family that Paper 8 compares against: uncalibrated critic spread, tens to hundreds of networks, and settings tuned per dataset. BCA seeks comparable selectivity from a conformally calibrated width on Bellman-residual size, with no extra critics, though unlike EDAC its current design has no completed D4RL results yet.

Paper 18: Conformal Prediction as Bayesian Quadrature (BQ-CP)

BQ-CP rereads conformal prediction as Bayesian numerical integration, so that a calibration set yields a whole distribution bounding a black-box model's expected loss instead of a single point estimate.

1. **What is the citation?** Snell, J. C., & Griffiths, T. L. (2025). Conformal prediction as Bayesian quadrature. In *Proceedings of the 42nd International Conference on Machine Learning* (Proceedings of Machine Learning Research, Vol. 267, pp. 56068–56084). PMLR. Version reviewed: arXiv:2502.13228v2. `https://doi.org/10.48550/arXiv.2502.13228`

2. **What is the research area?** The area is statistical machine learning, not RL: distribution-free uncertainty quantification (guarantees on a black-box model's errors that assume no particular data distribution). The paper re-derives split conformal prediction (a threshold set from held-out scores) and conformal risk control (its extension to other bounded, monotone losses). It derives both from Bayesian decision theory and Bayesian quadrature (numerical integration treated as inference, so the integral gets a posterior).

3. **What is the research question?** Can split conformal prediction and conformal risk control be recovered as special cases of a Bayesian procedure that needs no hand-picked prior? And can that procedure give guarantees conditioned on the calibration data actually observed, instead of averaged over calibration sets that were never drawn?

4. **What is the significance of the question?** Black-box predictors are used in high-stakes settings, and a deployment decision rests on the one calibration set in hand. Standard conformal guarantees are marginal: they hold on average over hypothetical calibration sets, so a particular set can still produce a threshold whose real risk is too high. Being frequentist, they also make it hard to use prior knowledge about deployment conditions.

5. **What are the contributions reported in this paper?** The paper makes four contributions. It frames split conformal prediction and conformal risk control decision-theoretically: both keep worst-case risk below a target α. It treats the expected loss as the area under the loss's quantile function (its value at each percentile), handled like Bayesian quadrature but without a prior. For this it takes the worst-case function through the observed losses and uses a classical result that gaps between the quantile levels of i.i.d. samples follow a flat Dirichlet distribution. It defines a random variable L^+ whose distribution upper-bounds the posterior of the expected loss, with both conformal methods recovered by using only its mean. It reports experiments with public code.

6. **What is the research method?** The method is theory followed by simulation. The appendix proofs yield a decision rule: sample the Dirichlet gaps 1,000 times and choose the smallest threshold λ for which L^+ stays below the target risk α with probability $\beta = 0.95$. The rule is tested over 10,000 random splits in each of three settings: a synthetic binomial loss (10 calibration points), a synthetic regression (200), and false-negative-rate control for multilabel classification on MS-COCO (1,000). The baselines are conformal risk control (CRC) and risk-controlling prediction sets (RCPS) built on a Hoeffding confidence bound.

7. **What are the results?** In theory, given the calibration losses, the expected loss is at most b with posterior probability at least $\Pr(L^+ \leq b)$ under any prior compatible with them. Empirically, CRC's threshold exceeds the target risk in 21.20%, 46.19%, and 45.05% of trials across the three settings; BQ-CP cuts this to 0.03%, 3.42%, and 5.43%, the last just above its 5% allowance.

RCPS records no failures but is widest (mean interval length 14.29 vs. 9.50 for BQ-CP and 7.99 for CRC; MS-COCO set size 3.57 vs. 3.04 and 2.92). At $n = 10$ BQ-CP is very cautious (mean risk 0.1758 against a 0.4 target).

8. **What are the impacts and limitations of the contributions?** BQ-CP shows that a conformal threshold can come with a distribution that upper-bounds the posterior of the expected loss, without a hand-chosen prior. It received an outstanding paper award at ICML 2025 and is the base WBCP (Paper 12) extends. The authors state that deployment data must be i.i.d. with the calibration data, losses need a known upper bound, and bounds valid for any prior can be loose. Beyond these, the advertised use of prior knowledge is left to future work, the rule returns one global threshold with no per-input guarantee, and no theorem bounds how often it fails over repeated calibration sets.

9. **How were the results evaluated or validated?** The results are validated by proofs and repeated sampling. The proofs assume i.i.d. continuous losses bounded by B; discrete losses, such as the binomial and 0–1 miscoverage losses of the two synthetic experiments, get only an unproved footnote. One step appears to take the Dirichlet law of the gaps, a sampling result, as their posterior once the losses are seen, without arguing this for every prior. The strengths are a synthetic loss with known true risk and 95% Clopper–Pearson intervals in two of three tables. The weaknesses are one real dataset, a single β, and an MS-COCO table with no intervals or stated α, whose 5.43% the text calls within the 5% limit.

10. **What are the ethical implications? How are they addressed?** A short impact statement says the method is meant to inform deployment of black-box models in safety-critical applications, that its guarantees may fail if its assumptions are violated, and that deployments should be monitored for distribution shift. Beyond the statement, a posterior bound can be mistaken for a safety certificate when the i.i.d. assumption goes unchecked. A bound on average loss can also hide high loss for a subgroup, though the introduction names fairness as a goal. The statement acknowledges the main risk but offers no monitoring method or subgroup analysis.

11. **What themes are suggested by this research? (minimum 3)** *The conservatism–improvement trade-off*: here it is caution vs. tightness, since RCPS records no failures but is widest, CRC is tightest but often exceeds the target, and BQ-CP sits between. *Uncertainty quantification and calibration*: a calibration set yields a distribution that upper-bounds the posterior of the expected loss, not one point estimate. *Theory vs. practice*: CRC's marginal guarantee can hold while, in the paper's three experiments, 21–46% of calibration sets give CRC a threshold whose risk exceeds the target.

Connection to the offline RL problem space and BCA: BQ-CP is in effect the threshold rule BCA runs: WBCP (Paper 12) with uniform weights is the exchangeable case that recovers it, and BCA sets the credibility level to this paper's $\beta = 0.95$. BCA inherits the guarantee only where the i.i.d. assumption holds: a posterior bound, given the calibration scores, on how often new scores from the same distribution exceed the threshold. This is neither a per-sample nor, as BCA's own documentation says, a policy-performance guarantee. Same-episode, policy-dependent Bellman residuals strain that assumption even after BCA thins its calibration set.

Part A: Literature Reviews

The three literature reviews are new in Version 2. Each is reviewed against the eleven literature-review questions, and their themes and meta-themes are drawn from the same four themes.

Literature Review 1: Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems (Levine et al., 2020)

This early tutorial explains why learning from a fixed dataset is hard, groups early methods by how they cope, and lists open problems.

1. **What is the citation?** Levine, S., Kumar, A., Tucker, G., & Fu, J. (2020). *Offline reinforcement learning: Tutorial, review, and perspectives on open problems* (arXiv:2005.01643v3) [Preprint]. arXiv. `https://doi.org/10.48550/arXiv.2005.01643`

2. **What is the survey scope?** The survey covers offline RL as a whole: learning a policy from a fixed dataset collected by a possibly unknown behavior policy, with no interaction during training. After RL basics it reviews three method families: importance-sampling estimators, which reweight logged data to mimic another policy; approximate dynamic programming (critic-based value learning) with policy constraints, uncertainty penalties, or conservative critics; and model-based methods. It then covers evaluation practice and applications such as robotics, healthcare, and driving. It is a tutorial and conceptual review for researchers entering the area, with roughly 190 references up to mid-2020. It sets aside exploration and a complete account of model-based RL.

3. **What is the survey question?** What prevents current RL algorithms from turning a fixed, previously collected dataset into the best policy the data can support, and which ideas have been proposed to overcome those obstacles?

4. **What is the significance of the survey?** It offered newcomers one entry point, placing older statistical work on importance sampling beside recent deep methods under one notation and one diagnosis: distribution shift created by counterfactual (what-if) queries. It appeared in May 2020, when offline RL had just become an active topic (BCQ and BEAR, Papers 13 and 14, and BRAC) but still went by competing names and lacked settled evaluation practice. Figueiredo Prudencio et al. later treat it as their main predecessor and present their open-problems section as an update of this one.

5. **What are the themes?** *Distribution shift*: this is the core obstacle, since learning offline means answering what-if questions about actions the data never contained. *The conservatism–improvement trade-off*: policy constraints keep the learned policy near the data, preferably within its support rather than matching it exactly, and conservative critics make the critic itself cautious. *Uncertainty quantification and calibration*: uncertainty-based methods make the critic cautious through an explicit uncertainty model.

6. **What is the survey method?** It is an expert narrative written as a tutorial, not a systematic review: it reports no search databases, keywords, or inclusion criteria. The material is organized by algorithmic principle rather than chronology, in one shared notation. The authors restate and explain results from the literature but prove no new theorems, run no experiments, and give no benchmark tables. Their own critical view appears mainly in the challenges subsections that close each method section.

7. **What are the survey findings?** Importance-sampling variance grows exponentially with the horizon, so those tools are most suitable when the learned policy stays close to the data. Naive dynamic programming can perform very poorly because its targets query unseen actions and the resulting errors compound uncorrected. Policy constraints help but must be tight, which can cap improvement. Uncertainty-based methods are principled, but calibrated uncertainty for deep value functions is hard to obtain, so constraint and conservative methods seemed to perform better at the time. Conservative critics can be too pessimistic, and how to adjust that pessimism on the fly is left open. The survey judges benchmarking immature.

8. **What are the limitations of the survey?** The main limitation is that it stops at mid-2020, before TD3+BC, IQL, ReBRAC, and the sequence-model methods. Coverage is uneven, with uncertainty-based methods given about half a page. No search protocol is reported. More than 30 of the roughly 190 references include one of the four authors, whose own methods (BEAR, BRAC, CQL) are among the families the survey reports as performing best. It offers no quantitative comparison and never treats offline hyperparameter selection directly.

9. **How was the survey assessed relative to other surveys?** It has no related-surveys section. It positions itself against the older batch RL literature and explains why it prefers the term *offline*. Of the three surveys reviewed here, it gives the most formal and conceptual account of *why* offline RL is hard. The survey by Figueiredo Prudencio et al. (2024) is broader and more systematic, with a taxonomy, post-2020 methods, and a benchmark review. He's survey (2023) is a short, more recent list of model-based methods with little analysis.

10. **What are the ethical implications? How are they addressed?** The survey has no ethics section. Safety appears only in scattered technical remarks, as a motivation rather than an ethical question. These remarks note that a half-trained treatment policy cannot be tried on real patients and that clinical records can over-represent severe cases and so mislead a naive learner. The survey does not address privacy and consent for clinical data, unequal treatment of groups under-represented in logs, or persuasive dialogue agents (one motivating example is a sales chatbot). Nor does it address the danger of treating off-policy estimates as the final check before deployment.

11. **What meta-themes are suggested by this survey? (minimum 3)** *Distribution shift*: the survey traces each family's main difficulty to the same mismatch, so later methods can be compared by where they intercept it. *The conservatism–improvement trade-off*: tight constraints are needed to keep errors small, yet they tie the policy to possibly poor behavior. *Uncertainty quantification and calibration*: this is a principled remedy that is hard to deliver, and dynamic modulation of conservative critics is left open.

Connection to the offline RL problem space and BCA: The survey lays out the method families of BCA's four host algorithms. BCA takes up the survey's open problem of how to dynamically modulate conservatism. Where the survey judged uncertainty estimates hard to calibrate, BCA uses residual widths calibrated on held-out transitions, though that calibration covers recorded transitions, not unseen actions.

Literature Review 2: A Survey on Offline Model-Based Reinforcement Learning (He, 2023)

This short primer on offline model-based RL profiles seven methods by how each builds caution into learning with a model of the environment.

1. **What is the citation?** He, H. (2023). *A survey on offline model-based reinforcement learning* (arXiv:2305.03360v1) [Preprint]. arXiv. `https://doi.org/10.48550/arXiv.2305.03360`
2. **What is the survey scope?** The survey covers offline model-based RL: fitting a dynamics and reward model to a fixed log of experience, then planning or optimizing a policy inside that model. After brief primers it profiles seven methods from 2020 to early 2023 (MOReL, MOPO, BREMEN, VI-LCB, COMBO, RAMBO, ARMOR) in about five pages, with nineteen references. Model-free methods appear only as a contrast. Benchmarks, quantitative results, off-policy evaluation, and application domains are absent.
3. **What is the survey question?** How have recent offline model-based methods coped with the distribution shift that arises when a model learned from logged data is used to optimize a new policy, and what problems remain unsolved?
4. **What is the significance of the survey?** He's paper is a dedicated, recent, and very short treatment of offline model-based RL that adds adversarial-model methods and sample-complexity theory. By May 2023, uncertainty-penalized models had been an established route to offline RL for about three years. Yet the two general surveys reviewed here gave model-based methods only a few pages and went no further than COMBO. The paper is a single-author preprint with no peer-reviewed version listed on arXiv, so it serves as a quick map rather than an authoritative reference.
5. **What are the themes?** *Distribution shift*: the survey calls this the main problem, made worse by model exploitation; a policy optimized inside a model learned from logged data may be out of distribution at deployment and do worse. *The conservatism–improvement trade-off*: COMBO's conservative critic and the adversarially trained models of RAMBO and ARMOR supply caution by means other than uncertainty penalties. *Uncertainty quantification and calibration*: MOReL and MOPO turn ensemble uncertainty into a reward penalty.
6. **What is the survey method?** It is a personal, narrative selection: no databases, search terms, or inclusion criteria are named, and the choice of these seven methods is not explained. Each profile gives the core idea, the key equations in the notation of whichever source it follows, and the original authors' claims. There are no figures, tables, experiments, new results, or unifying notation. Some passages follow Figueiredo Prudencio et al. closely.
7. **What are the survey findings?** By the survey's account, distribution shift, made worse by model exploitation, remains the defining open problem, and none of the reviewed methods provides a performance guarantee that holds regardless of the data. Reward-penalty methods can reuse existing planners, but they depend on uncertainty estimates the survey regards as error-prone, a dependence COMBO avoids. The one optimality result it reports, for a penalized form of value iteration, is confined to tabular problems. Citing the ARMOR paper, it adds that many theoretically grounded methods go unused because they fail to beat the policy already in service.
8. **What are the limitations of the survey?** Coverage is thin (seven methods, nineteen references), with no benchmark results, comparison table, or stated selection procedure. Most

profiles restate equations and pass on the authors' claims with little independent critique. Some descriptions are loose: COMBO is said to penalize rewards, while the objective the survey prints acts on Q-values (the critic's action values). It has not been revised since its first version.

9. **How was the survey assessed relative to other surveys?** It cites the other two surveys only as background authorities and never says what it adds to them. Of the three reviewed here it is the narrowest, shortest, and most recently written. Levine et al. give model-based methods a more analytical section, and Figueiredo Prudencio et al. place them in a full taxonomy with benchmark context. He's distinct additions are RAMBO, ARMOR, VI-LCB, and the tabular sample-complexity result, none of which appears in either broader survey.

10. **What are the ethical implications? How are they addressed?** The survey discusses none. It frames offline learning as improving on a deployed system from that system's own data. In that setting, model exploitation means a policy can look excellent inside the learned simulator and fail once it replaces that system. Biases in past decisions can pass into both the model and the new policy. With off-policy evaluation omitted, the survey never explains how such policies could be checked before release. Only the technical safe-improvement guarantees of COMBO and ARMOR come near the topic.

11. **What meta-themes are suggested by this survey? (minimum 3)** *The conservatism–improvement trade-off*: several methods vary their caution across state-action pairs, but where the survey says how its strength is set, it is a hand-chosen hyperparameter. *Uncertainty quantification and calibration*: MOReL's and MOPO's penalties rest on ensemble heuristics whose link to actual model error the survey does not examine. *Theory vs. practice*: the only optimality result the survey reports is tabular, and it relays that many provable methods go undeployed.

Connection to the offline RL problem space and BCA: He's survey suggests the model-based branch, like BCA's model-free hosts, sets its caution by hand-chosen hyperparameters and sizes some penalties by uncertainty estimates it calls possibly inaccurate. Because a method like COMBO mirrors CQL's penalty, BCA's per-sample widths might carry over to this branch, which is untested.

Literature Review 3: A Survey on Offline Reinforcement Learning: Taxonomy, Review, and Open Problems (Figueiredo Prudencio et al., 2024)

This broad, structured update to Levine et al. offers a taxonomy that describes each offline algorithm as a class plus optional loss terms, a review of benchmarks, and a refreshed list of open problems.

1. **What is the citation?** Figueiredo Prudencio, R., Maximo, M. R. O. A., & Colombini, E. L. (2024). A survey on offline reinforcement learning: Taxonomy, review, and open problems. *IEEE Transactions on Neural Networks and Learning Systems, 35*(8), 10237–10257. `https://doi.org/10.1109/TNNLS.2023.3250269`

2. **What is the survey scope?** The survey covers deep offline RL in the single-agent setting: direct and implicit policy constraints, importance sampling, regularization, uncertainty estimation, model-based methods, one-step methods, imitation learning, and trajectory optimization. Its timeline runs from late 2018 to early 2022, its method table classifies 26 entries, and it cites 91 references. It also reviews off-policy evaluation and the D4RL and RL Unplugged benchmarks. It leaves out theoretical analyses, multi-agent methods, and most marginalized importance sampling.

3. **What is the survey question?** How can the fast-growing body of offline RL algorithms be placed in one framework that shows each method's design choices, how those choices fare on different kinds of data, and where research effort should go next?

4. **What is the significance of the survey?** The survey supplies an updated review in a single notation and a taxonomy that lets methods be read as combinations of reusable parts. It also maps dataset properties to the algorithm classes that handle them, which is meant to help practitioners match methods to their data. When it appeared on arXiv in March 2022, the main broad review was the 2020 tutorial. Many newer methods (TD3+BC, IQL, COMBO, the Decision Transformer) had since been published, several relying on strategies the tutorial never discussed.

5. **What are the themes?** *Distribution shift*: most classes are introduced as ways to prevent, penalize, or sidestep this central difficulty. *The conservatism–improvement trade-off*: this appears as policy constraints and behavior regularization, with direct constraints that estimate the behavior policy set against implicit ones that use only dataset samples. *Theory vs. practice*: this concerns evaluating policies without deployment, with off-policy evaluation tied to hyperparameter selection and early stopping.

6. **What is the survey method?** It is an expert survey whose selection process is not reported: it names no databases, search strings, or inclusion criteria, and the authors note that their method table is far from exhaustive. The taxonomy is the organizing device, with every method restated in a shared notation and a timeline ordered by preprint date. The only quantitative analysis is secondary: a heatmap built from D4RL scores collected from the literature, rescaled per dataset and averaged by dataset property. No new experiments are run.

7. **What are the survey findings?** Most new algorithms are different responses to distribution shift, each with a trade-off. Policy constraints can be overly pessimistic. Direct ones also need a good estimate of the behavior policy, while implicit ones avoid that estimate and suit online fine-tuning better. The survey relays two lessons: hyperparameter search mattered more than the choice of divergence (from BRAC), and performance is often not tied to algorithmic complexity (from TD3+BC). It calls model-based uncertainty estimation difficult and often unreliable. Its open problems are led by reliable off-policy evaluation and hyperparameter tuning, then incremental RL for changing data and fine-tuning, safety-critical RL, and use of unlabeled data.

8. **What are the limitations of the survey?** No selection rules are stated, and depth is uneven: the uncertainty section reviews only one line of ensemble work. The introduction promises a discussion of dynamically adjusted conservatism that the open-problems section never delivers. The heatmap pools scores from many papers, so it likely mixes tuning budgets and codebases; it also shows no variance and credits each class with its best member. The algorithmic review stops at early 2022, so later methods such as ReBRAC are absent. At least 21 of the 91 references share a single coauthor from another research group, which may tilt the review toward that group.

9. **How was the survey assessed relative to other surveys?** It positions itself against one predecessor, Levine et al. (2020), and claims to add a taxonomy, a single notation, the methods published since, a benchmark review, and a performance figure. Of the three surveys reviewed here it gives the broadest map of model-free methods and is the only one to systematically map benchmark environments to dataset properties. The tutorial by Levine et al. remains longer

and more conceptual. He's preprint is far narrower but reaches theory-driven and adversarial model-based work that this survey omits.

10. **What are the ethical implications? How are they addressed?** The survey has no ethics section. Safety appears as a motivation (testing a policy outside the data's support could be hazardous in driving or medical diagnosis) and as an open problem (safety-critical RL). The survey asks for datasets with risky gaps, such as driving logs without a single crash, to test whether methods learn to avoid hazards missing from the data. It does not discuss prejudice carried in logged human decisions, privacy, consent, or accountability. As scholarship it discloses funding and avoids self-citation.

11. **What meta-themes are suggested by this survey? (minimum 3)** *The conservatism–improvement trade-off*: in its equations most mechanisms carry a single scalar weight or threshold, and dynamic modulation of conservatism is named as an open problem but not developed. *Uncertainty quantification and calibration*: the survey calls model uncertainty estimates often unreliable but does not discuss calibration (whether an estimate matches observed error). *Theory vs. practice*: off-policy evaluation and hyperparameter selection lead the list of open problems.

Connection to the offline RL problem space and BCA: This survey explains three of BCA's four host algorithms side by side (CQL, TD3+BC, IQL). It names, without developing it, the open problem BCA targets: dynamically modulating the degree of conservatism. Its reliance on D4RL is also a caveat for BCA, since any results measured on that benchmark alone carry its limits.

Part B: Themes and Observations

All twenty-one works (eighteen research papers and three literature reviews) are described with the same four themes. Each paper names three, and for a review I count its themes and meta-themes together.

Theory vs. practice is the most common (19 works). What is proved or reported often differs from what an offline user gets: in several papers the guarantees assume conditions the tested algorithm does not meet, and reported scores depend on implementation details and on tuning with live trials.

The conservatism–improvement trade-off comes next (18 works). An offline method has to decide how far to distrust values its data cannot verify. Too little caution lets errors spread, and too much ties the policy to the logged behavior. Many methods set that caution with one hand-chosen number, which UNIQ argues is too careful where data are dense and may be too lax where they are sparse.

Distribution shift (16 works) is the problem the other themes respond to: values for actions the data do not contain cannot be corrected offline.

Uncertainty quantification and calibration is the least common theme (12 works), but it is the one most relevant to this project, because it offers a way to make caution vary per sample. The three surveys describe uncertainty estimates as hard to obtain or often unreliable, and QQL, UWAC, and EDAC act on uncertainty signals that carry no calibration guarantee. Five works add a calibration step (Papers 8, 10, 11, 12, and 18), and this review notes limits on each guarantee.

These themes motivate BCA and also bound it. BCA is designed to set caution per sample using calibrated uncertainty, but a calibration guarantee is not a policy-performance guarantee, and BCA's benefit has not yet been shown on benchmarks.